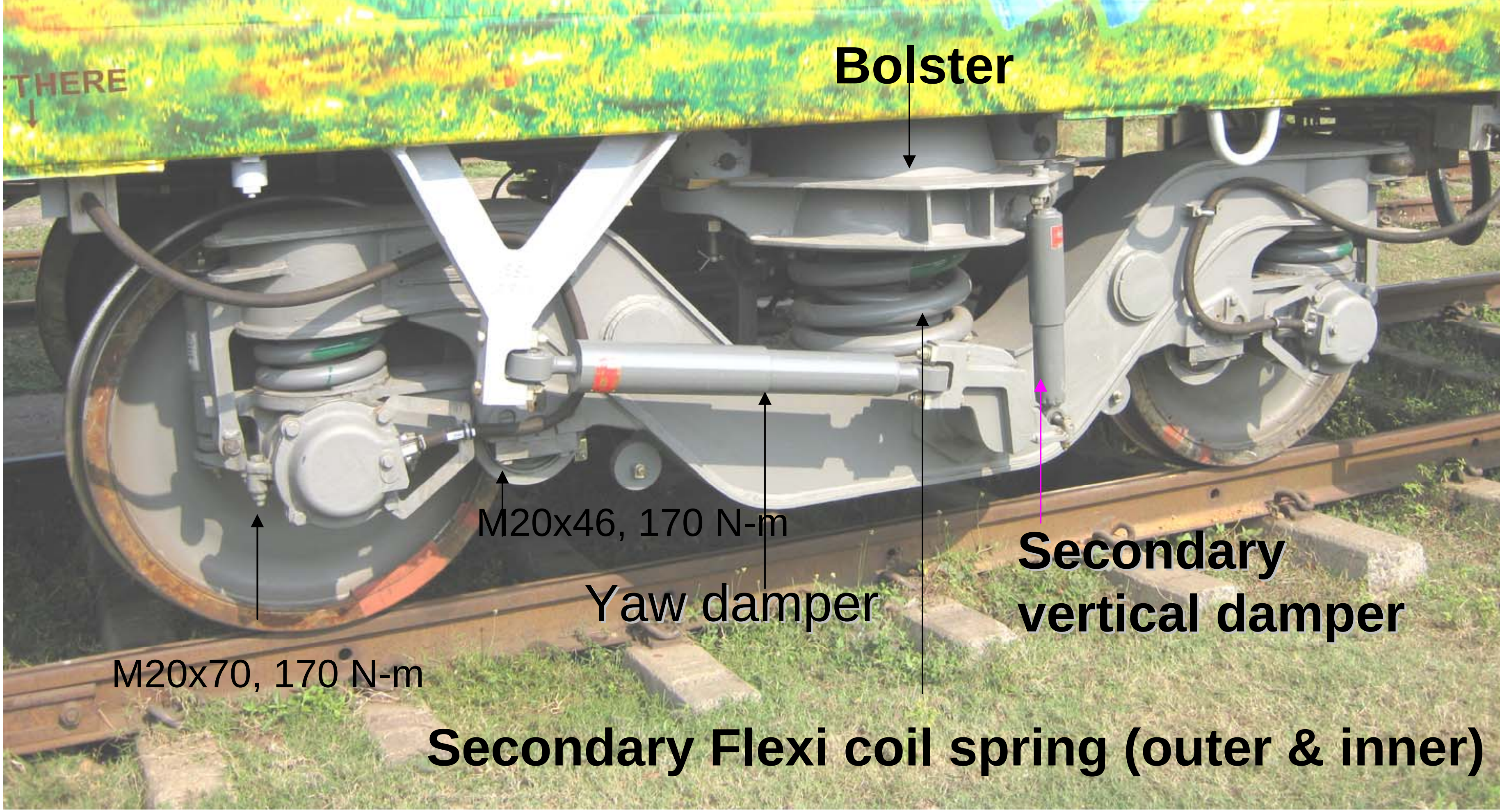
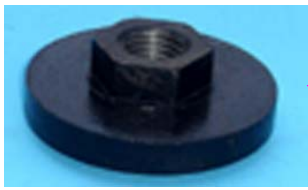
Presentation for LHB coach



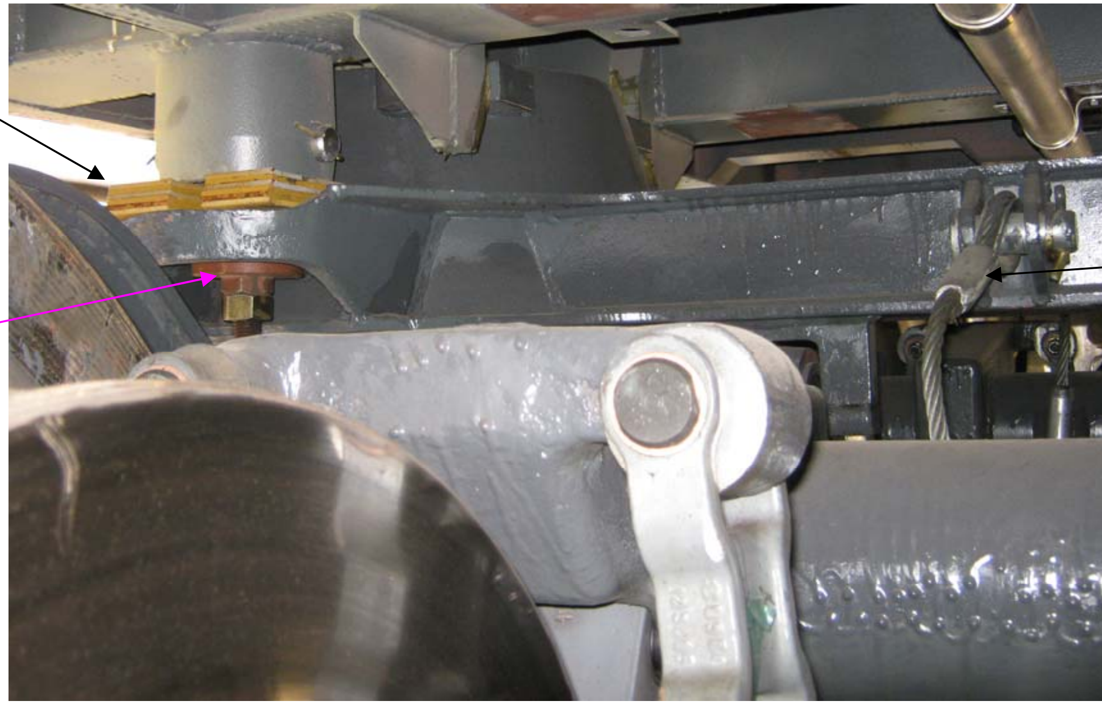
FIAT Bogie of LHB coach

BUFFER HEIGHT ADJUSTMENT

Shims



Swing nut



Safety wire rope connected between Bolster & bogie cross beam

- Shims at body/bogie connection.
- Shims NOT added/removed in Primary and Secondary Suspension for wheel wear compensation or buffer height adjustment.


Link pin
Shim
Swing link pin connecting body to bogie

Maintenance of Dampers

Proper torquing to be ensured

Primary vertical damper		100 Nm
Secondary vertical damper	M 12X100	70Nm
Secondary lateral damper	M 12X55	70Nm
Yaw damper	M 16X120	170Nm

Use proper corrosion resistant chemicals (WD-40) on threads left open after assembly

Tightening torques

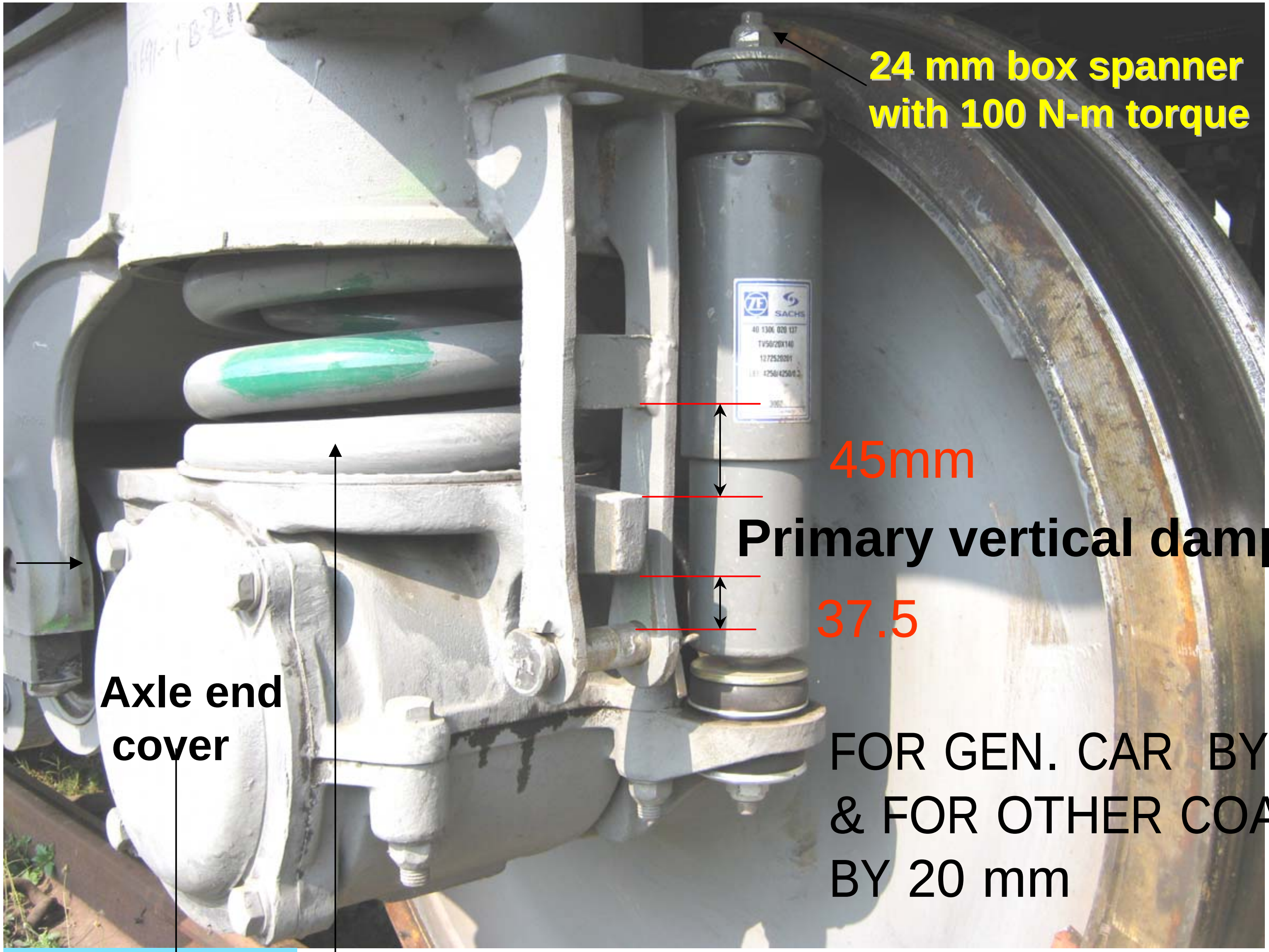
Screw type	Tightening torque
M 8	21 Nm
M 10	40 Nm
M 12	70 Nm
M 16	170 Nm
M 20	340 Nm
M 24	590 Nm
M 30	1170 Nm

Rubber Spring

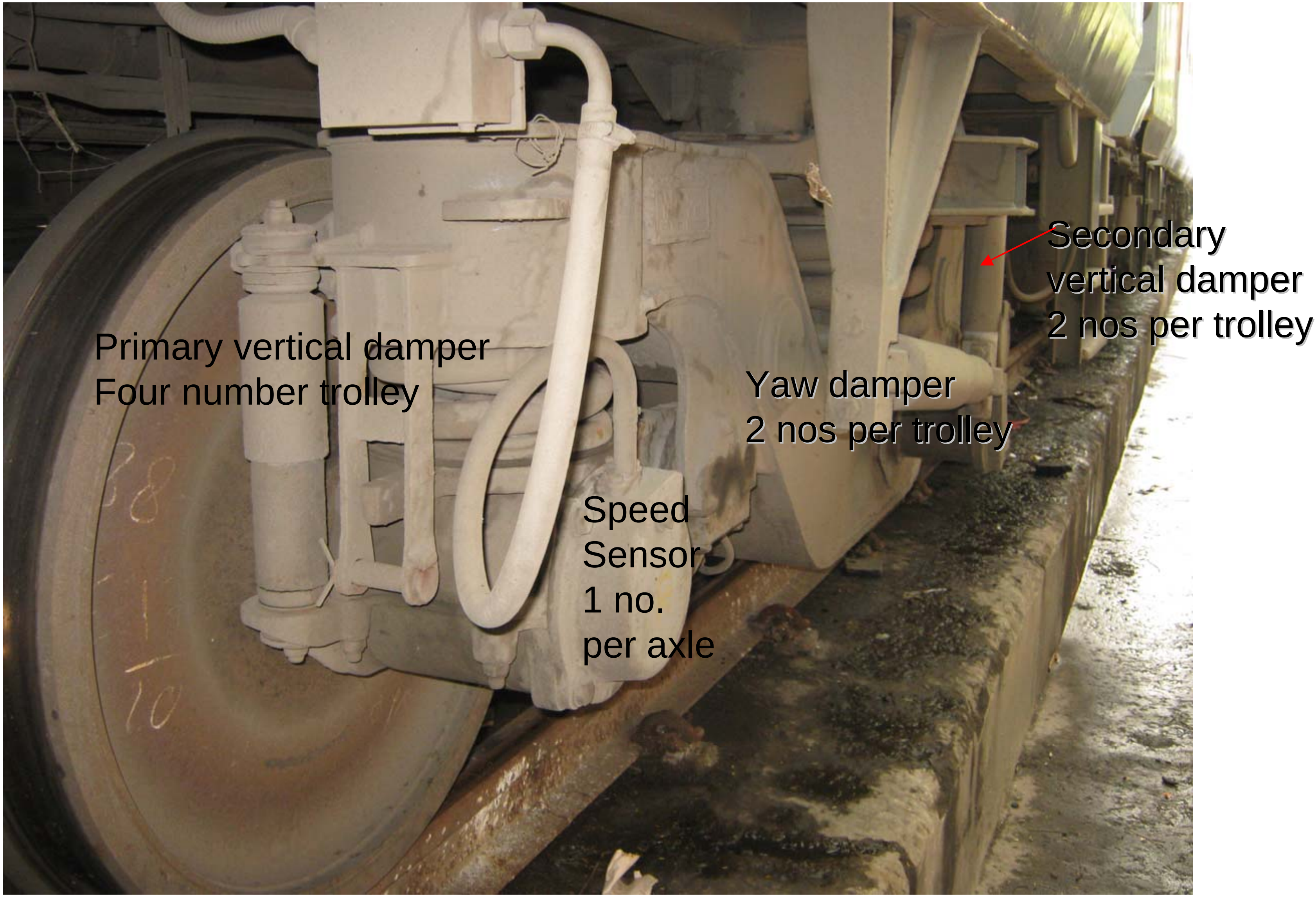
Bolster Flexi coil spring inner & outer

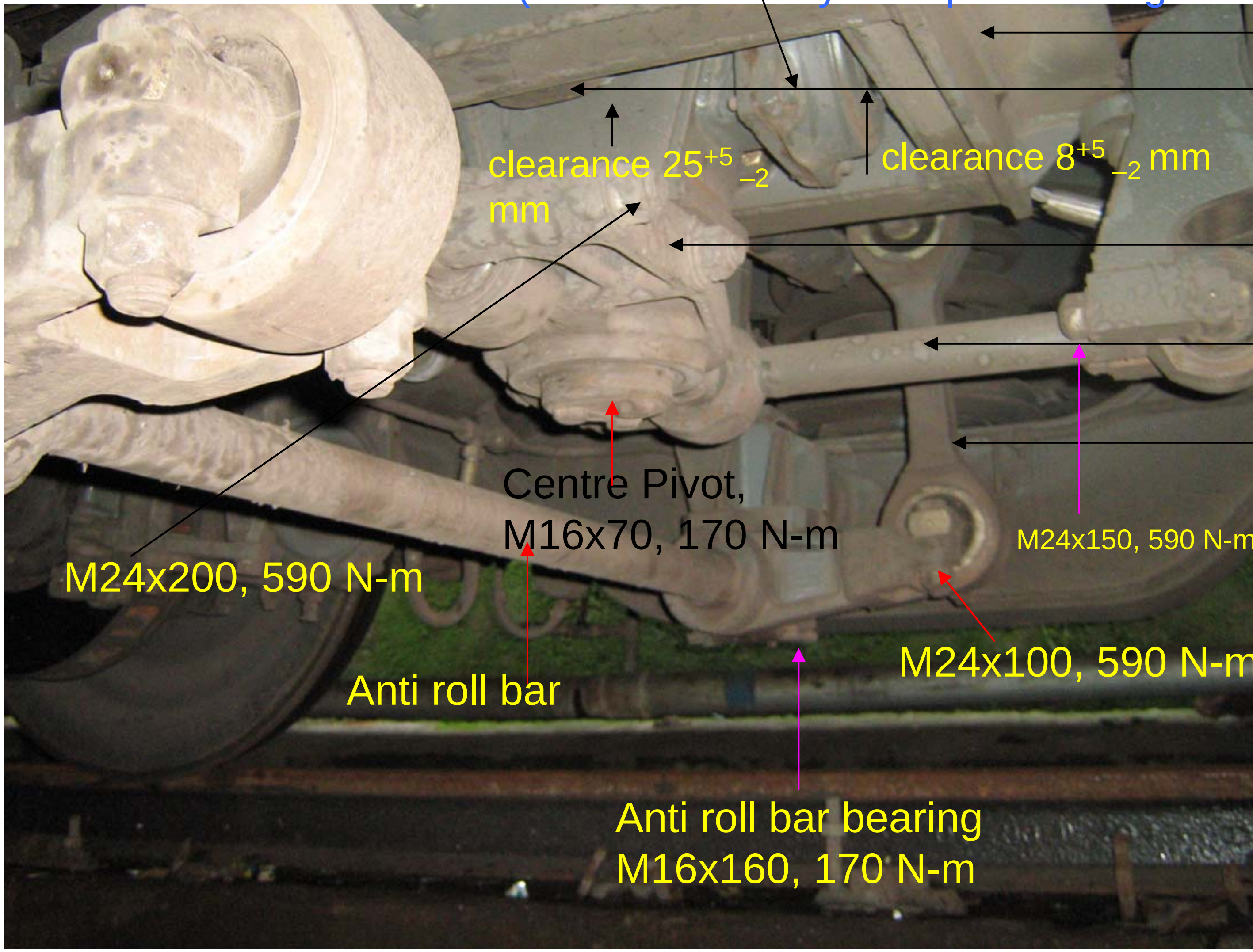
Coil-to-coil impression in Weak flexi coil bolster spring

M16x45,
160 N-m



Primary spring outer & inner


Primary vertical damper
Four number trolley
Speed
Sensor
1 no.
per axle
Yaw damper
2 nos per trolley
Secondary
vertical damper
2 nos per trolley



Traction lever

Anti roll bar assembly

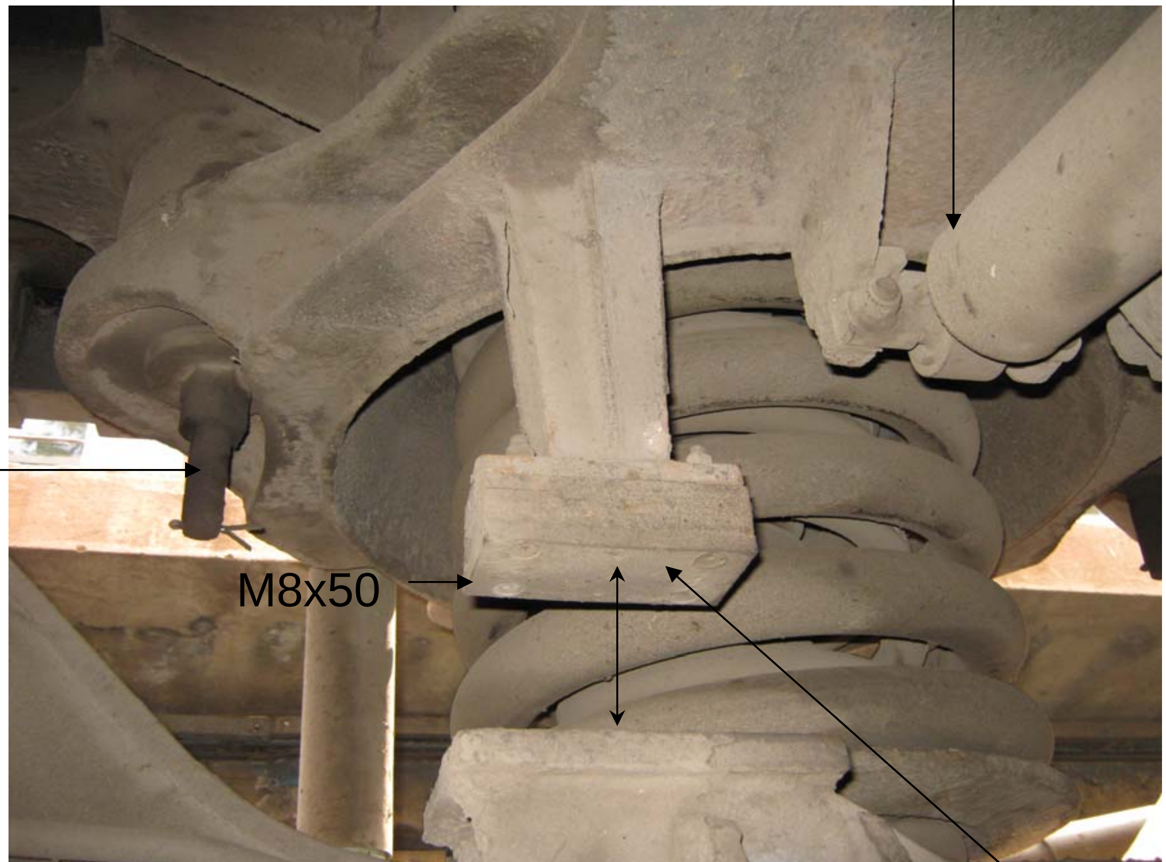
It is a torsion bar with two forks- between bogie frame & bolster, connected by roll links.

It resists rolling motion of coach.

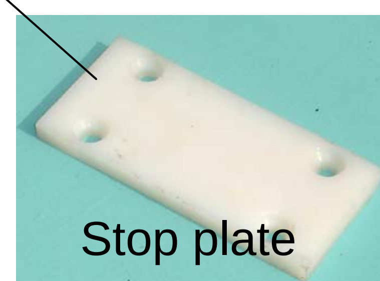
Lateral damper (1 no. for each trolley)

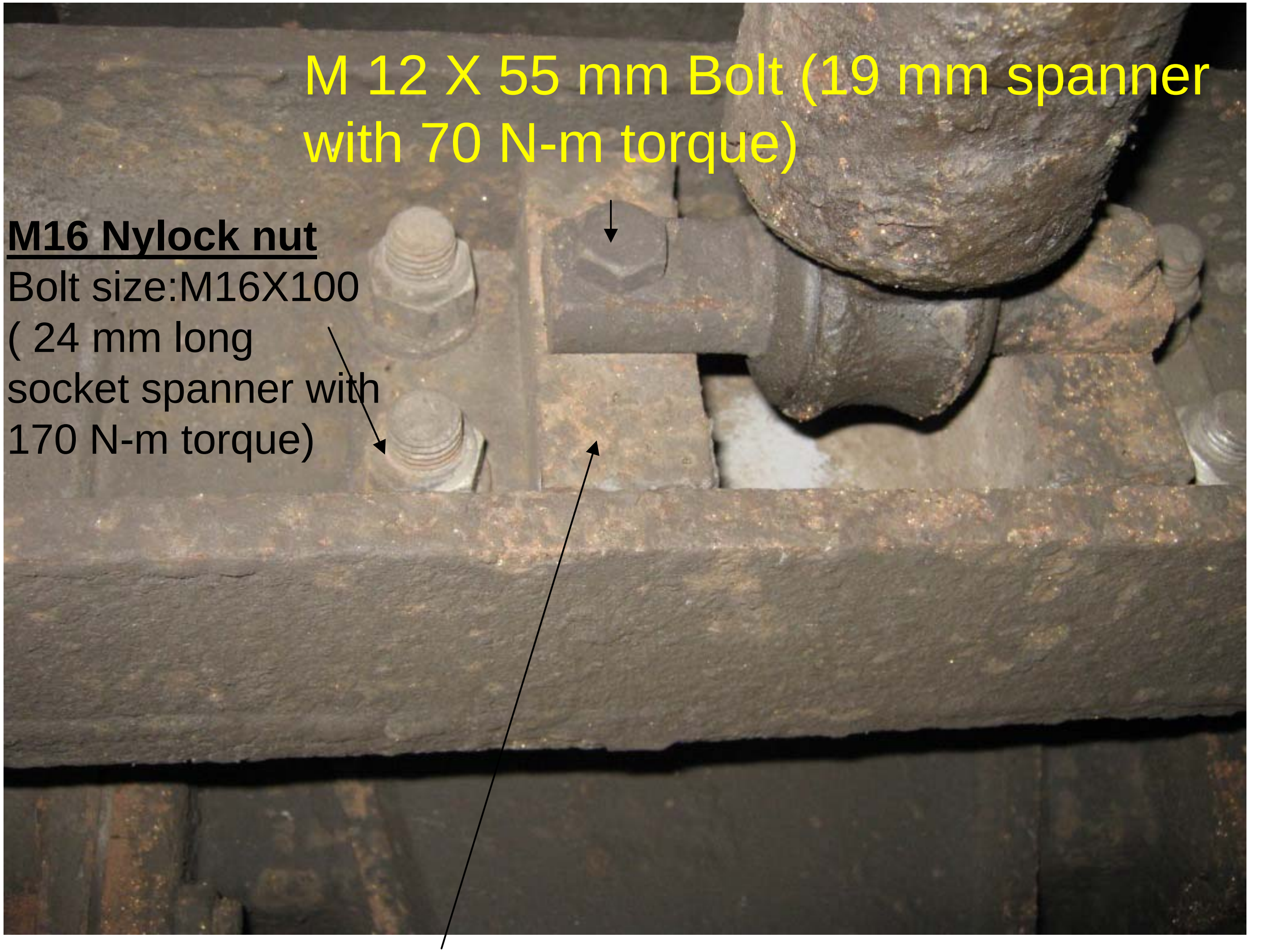
Swing link pin
(Threaded pin)

(46 mm spanner
spanner is
required,250 N-m
torque is required)



Secondary bump stop gap 95 $^{+0}/_{-5}$ mm



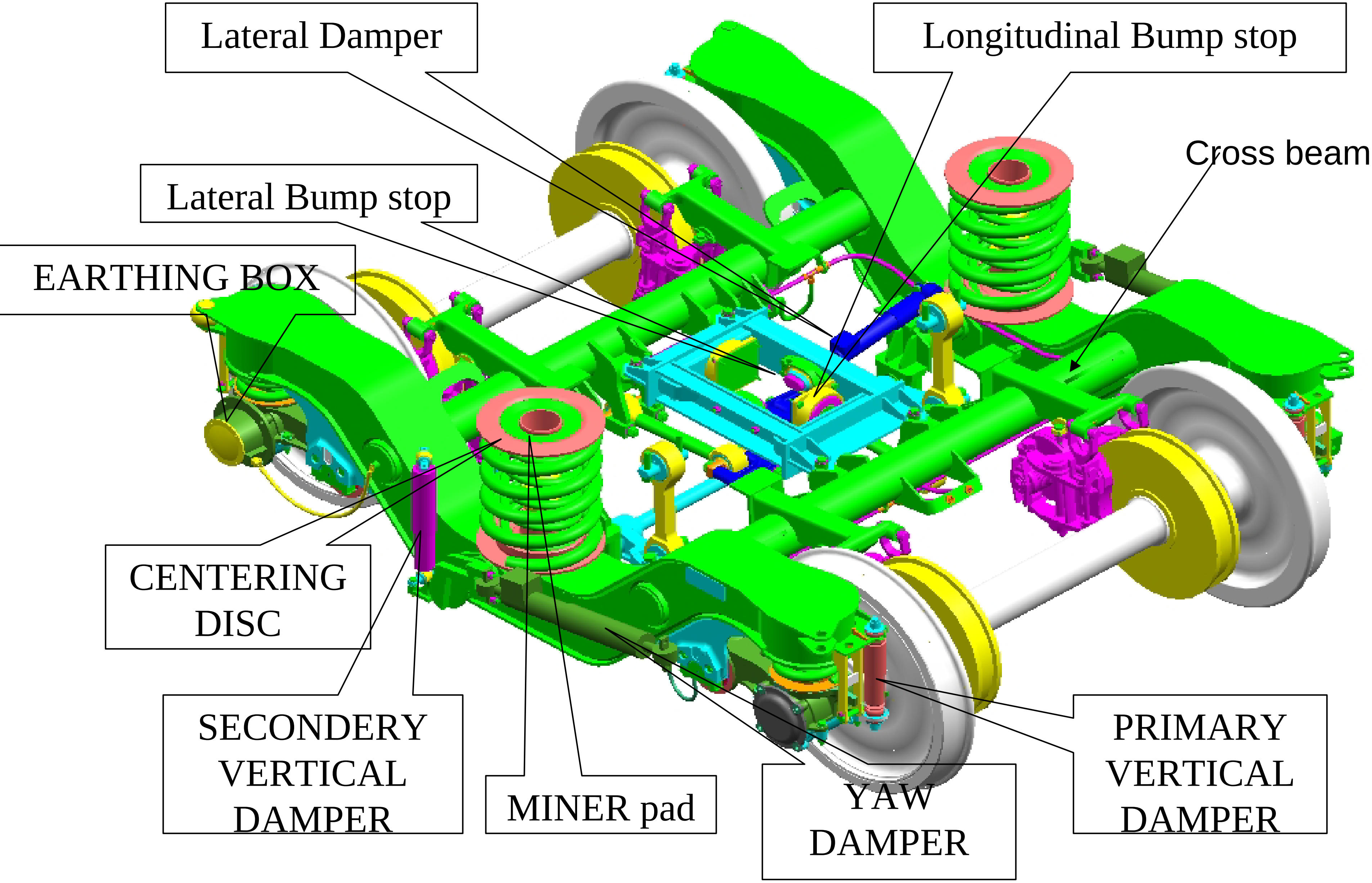


Lateral damper bracket
(fitted with lateral bump stop)

Lateral damper bracket

Lateral bump stop

(Bolster removed)



Principles of force transmission

Vertical forces:

⌘Body-bolster-miner pad-Sec. suspension-Bogie frame-Primary springs/Control arm-Axles

Lateral forces:

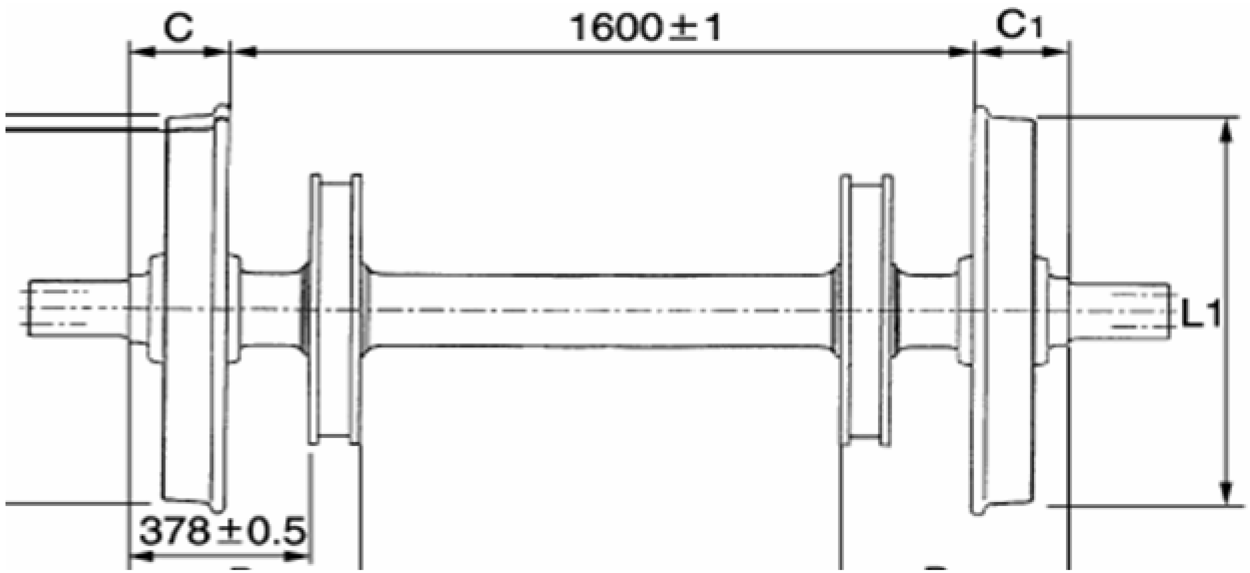
Body-Bolster-miner pad/Sec. springs/lateral bump stop-Bogie frame-Control arm-Axles

Traction and braking forces:

⌘Body-Traction Centre-Traction Lever/longitudinal Bump Stop-Bogie Frame-Control Arm-Axles.

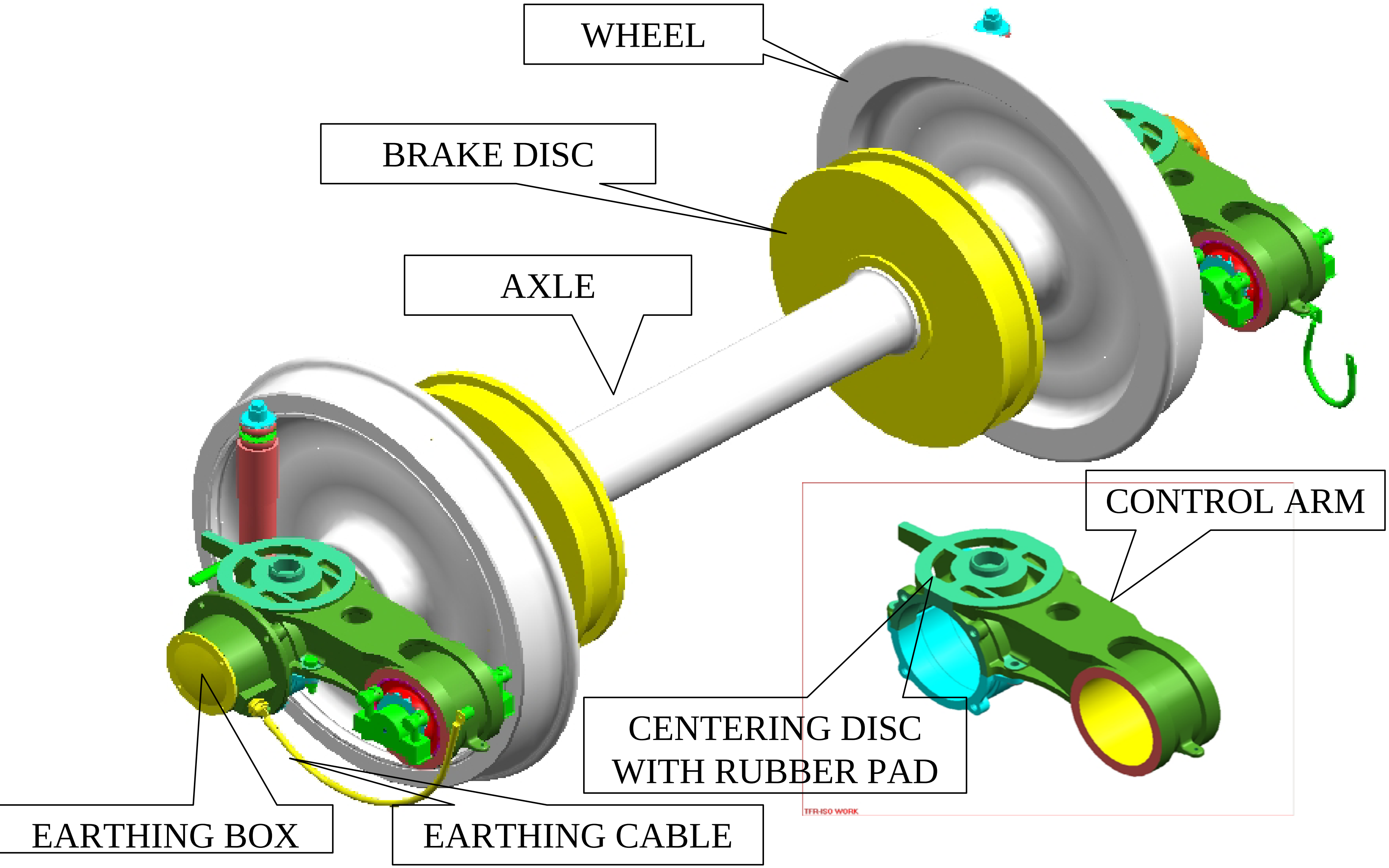
Running gear
in LHB coach

WHEEL SET:



- Two brake disks (4), diameter 640 mm and width 110 mm.
- Two wheel discs of dia 915 (New), 845 (worn).

(Wheel set with Primary springs removed)



Control Arm top

Control lower left

Control lower right

- CARTRIDGE TAPER ROLLER BEARING

Two manufacturer

- Timken
- SKF

Shelled tread wheel

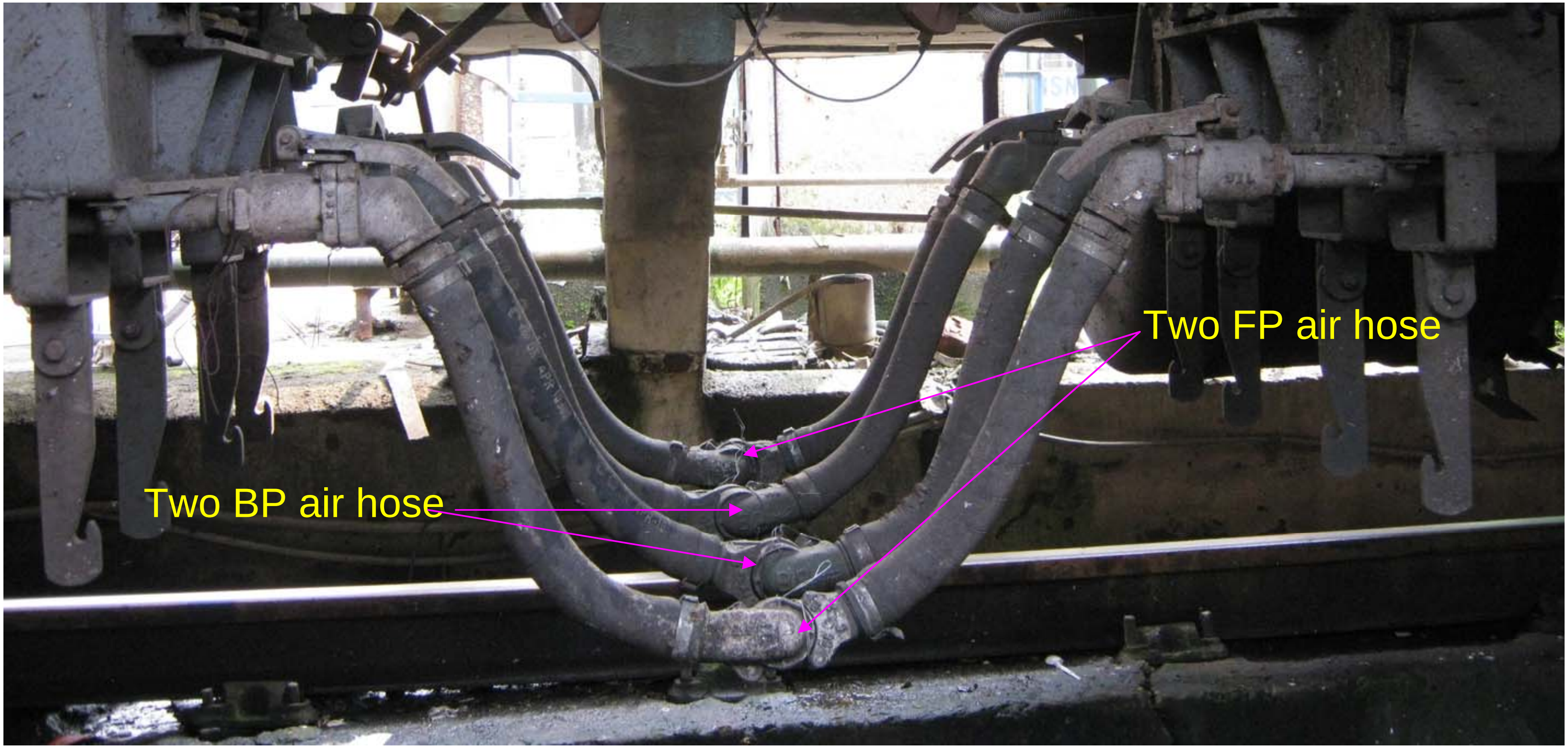
Spaded rim as well grooved wheel

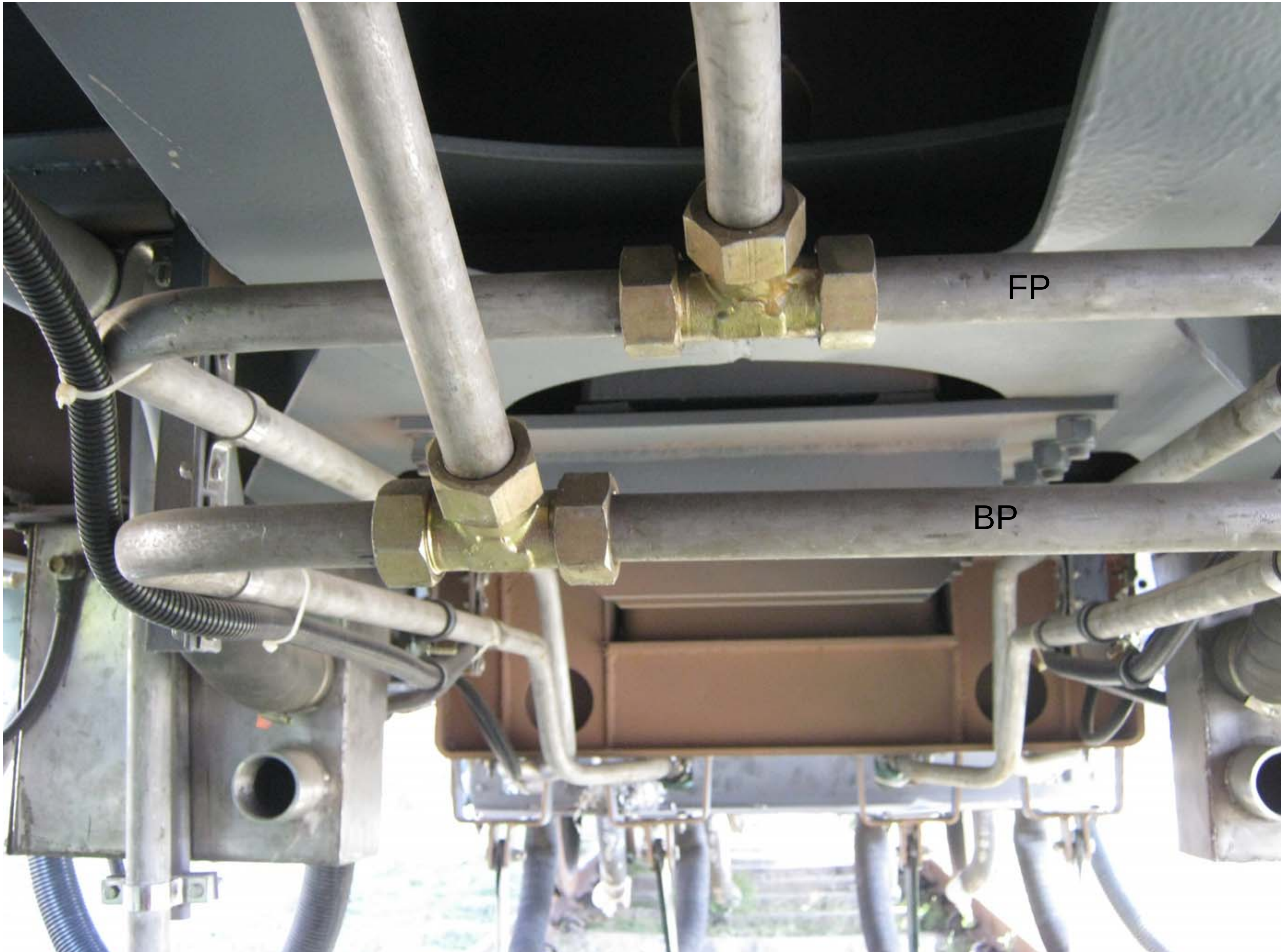
- Zonal Railways have reported low life of wheels due to wheel shelling. This problem is currently under study with RDSO.
- Following action has been taken/ suggested by RDSO to Rlys. :
  - Brake cylinder pressure reduced to 3 bar from 3.8 bar
  - Material of wheel upgraded as per RDSO spec R-19 rev 3. Fitment started with new specification since April 08

Air brake system in LHB coach

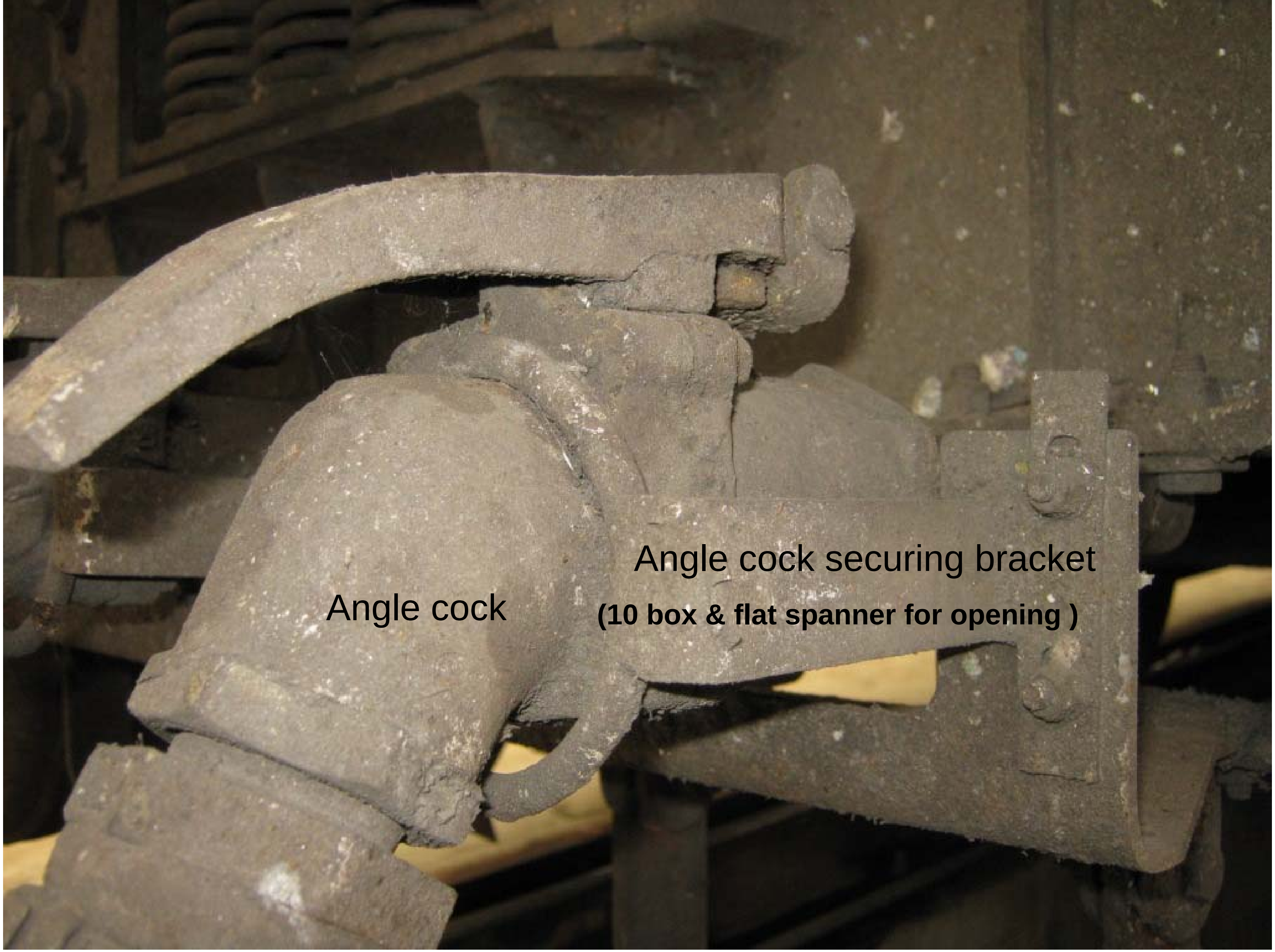
There are two manufacturer:

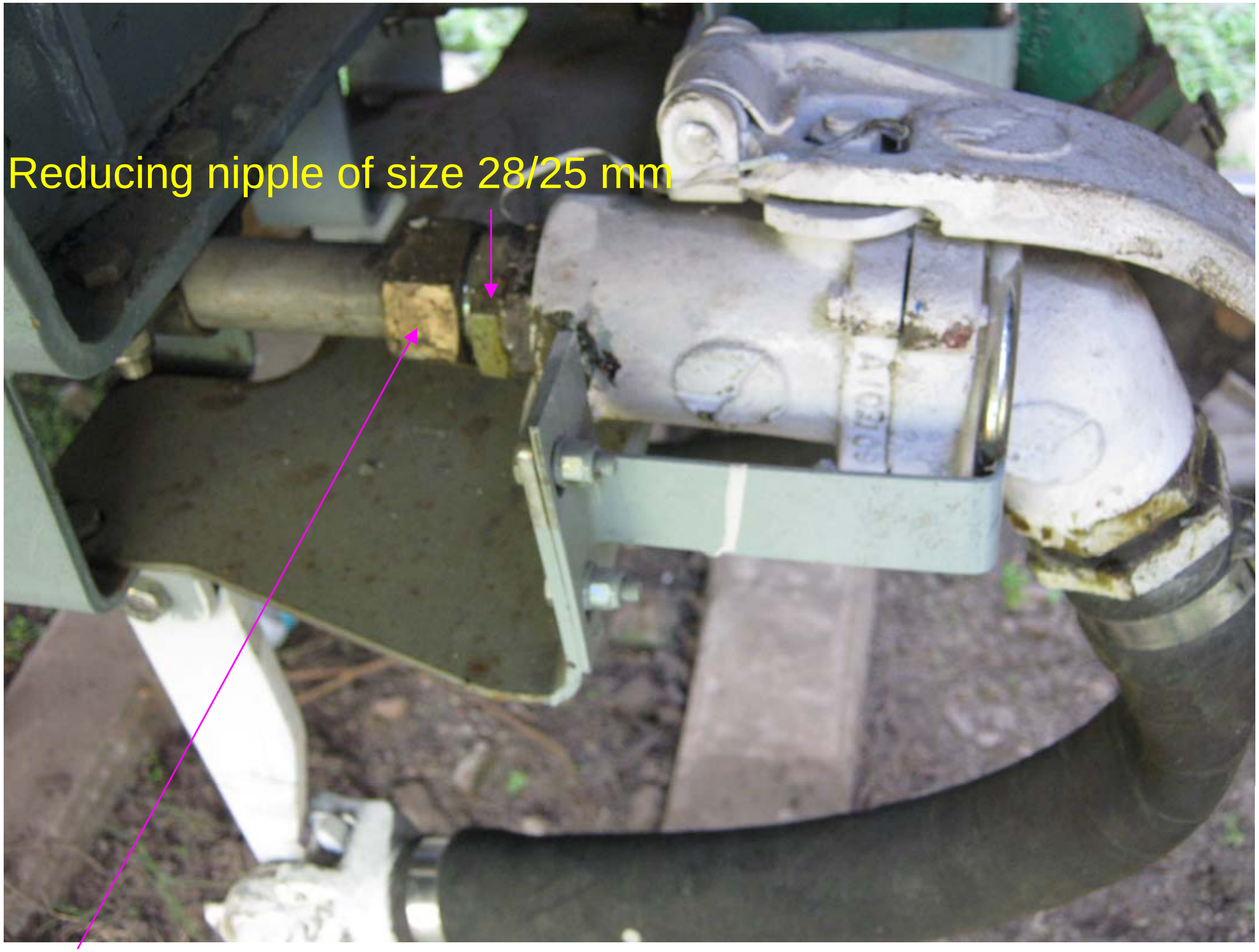
1. Knorr Bremse (KB)
2. Faiveley Trannsport India Limited (FTIL)


Two FP air hose
Two BP air hose

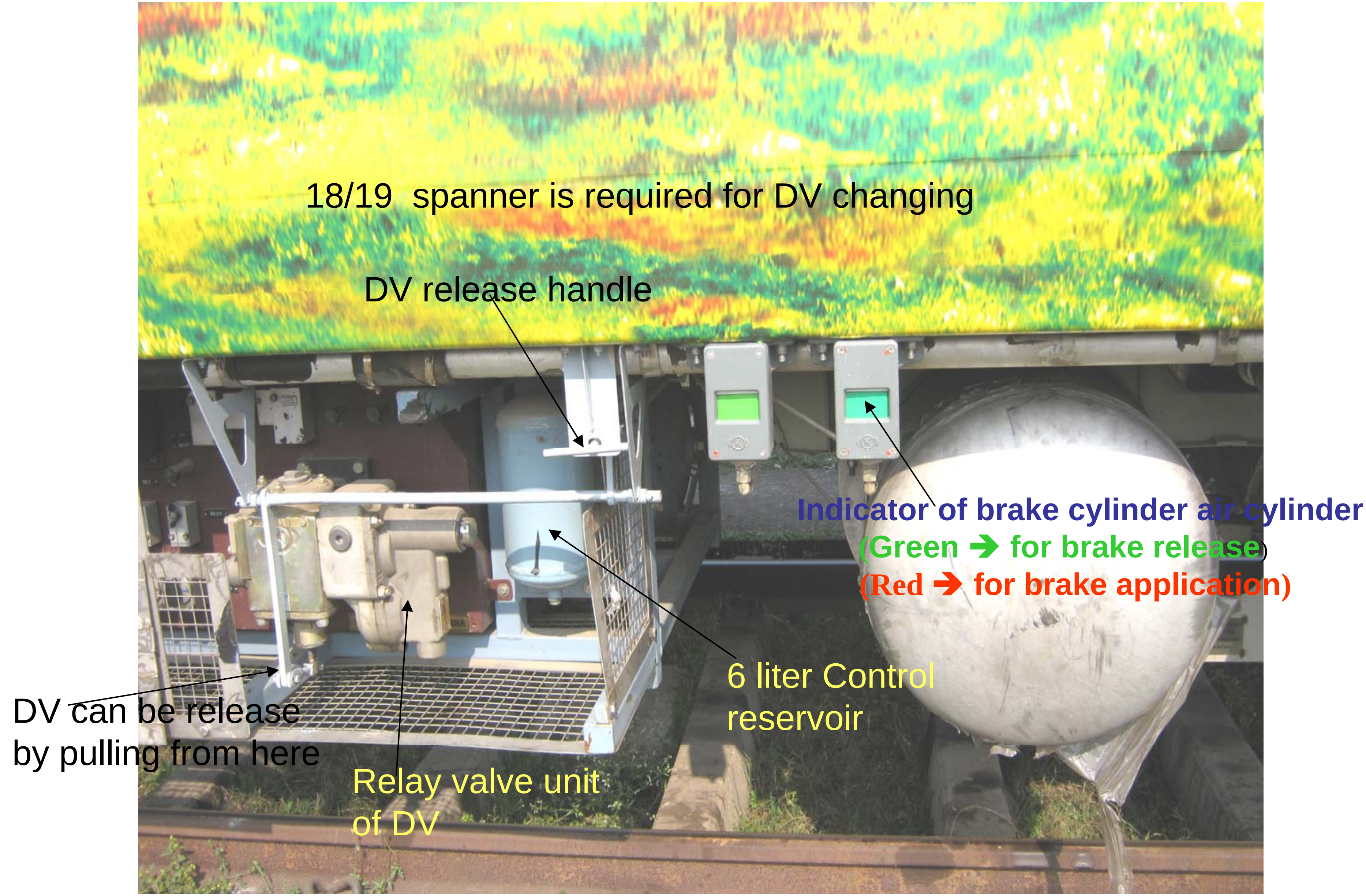


BP FP pipeline arrangement


Angle cock securing bracket
(10 box & flat spanner for opening)
Angle cock



Ferrule joint in angle cock

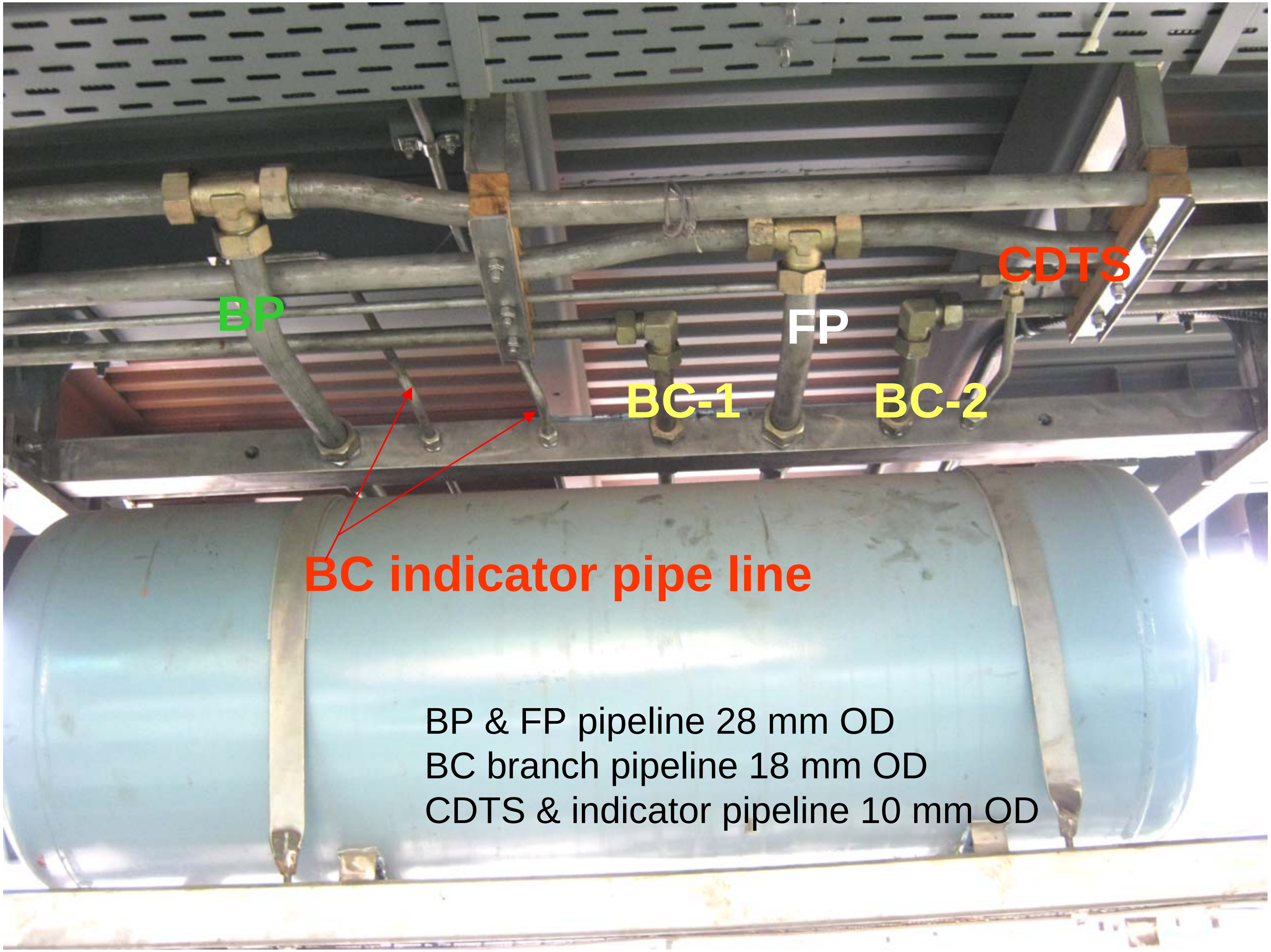


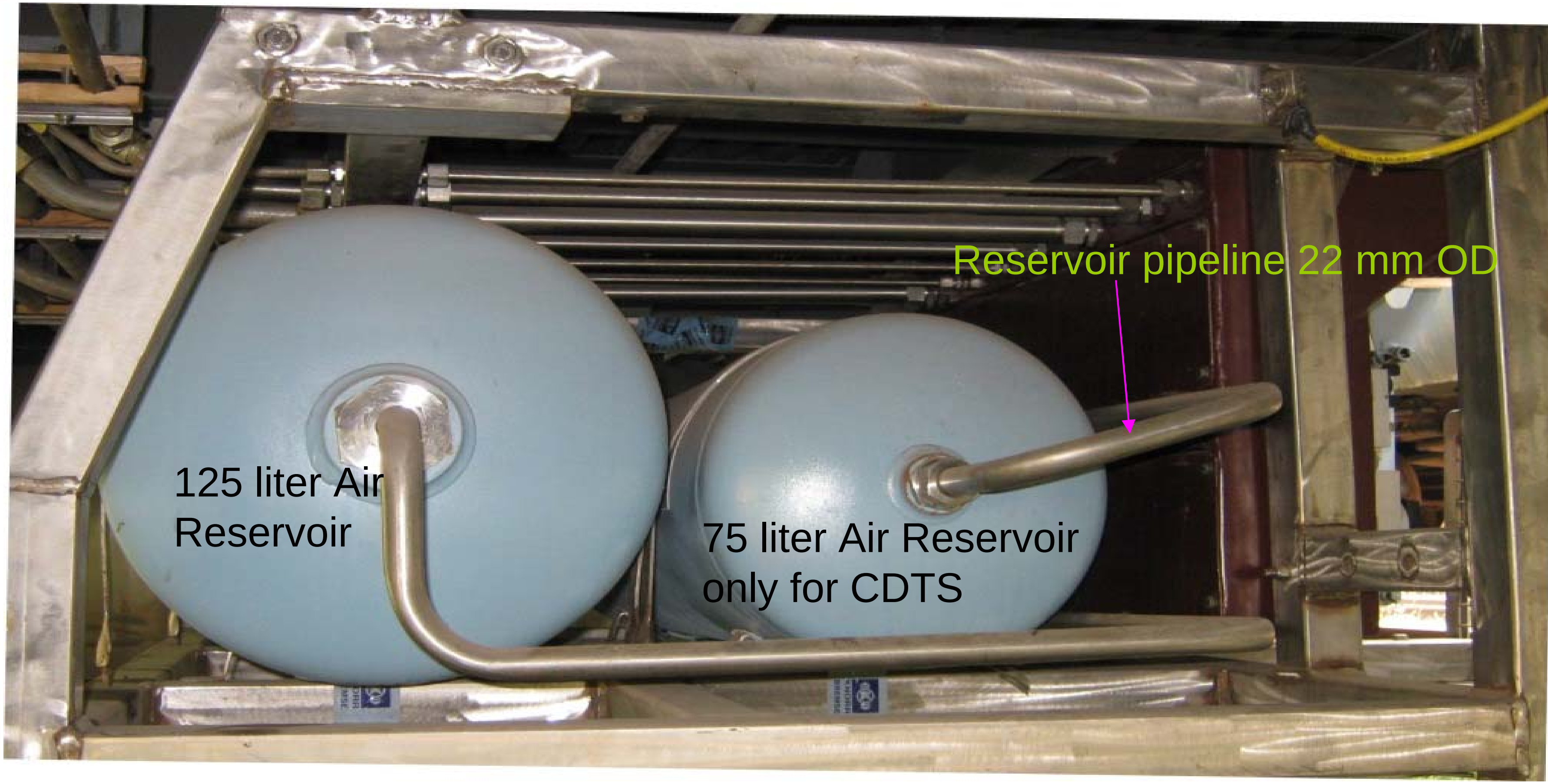
Knorr Bremse Make Brake Container (Brake Equipment Panel)

FTIL make Brake Container (Brake Equipment Panel)

FTIL make Brake Container (Brake Equipment Panel)

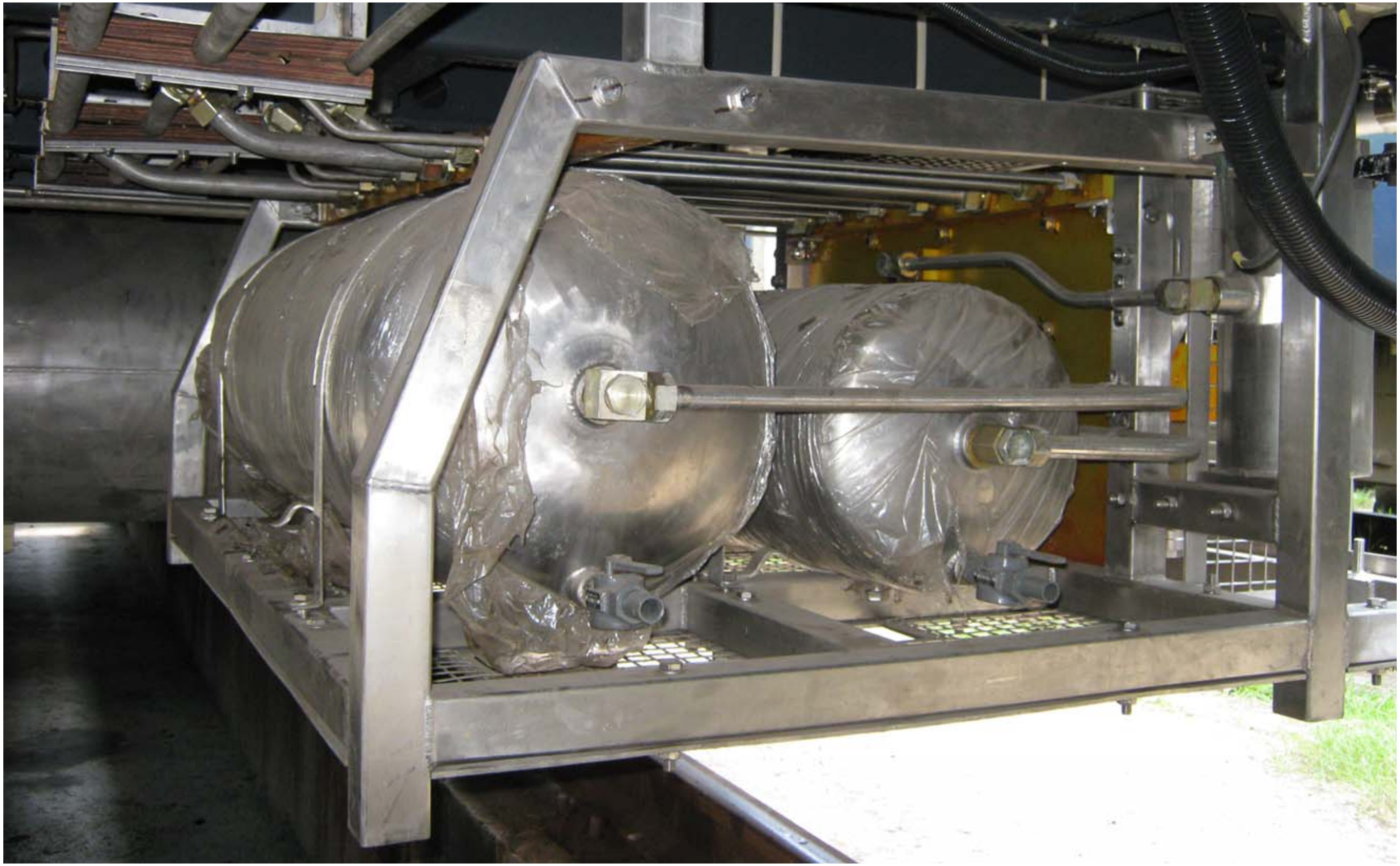
KB make Isolating cock in open position


BP
FP
CDTS
BC-1
BC-2
BC indicator pipe line
BP & FP pipeline 28 mm OD
BC branch pipeline 18 mm OD
CDTS & indicator pipeline 10 mm OD

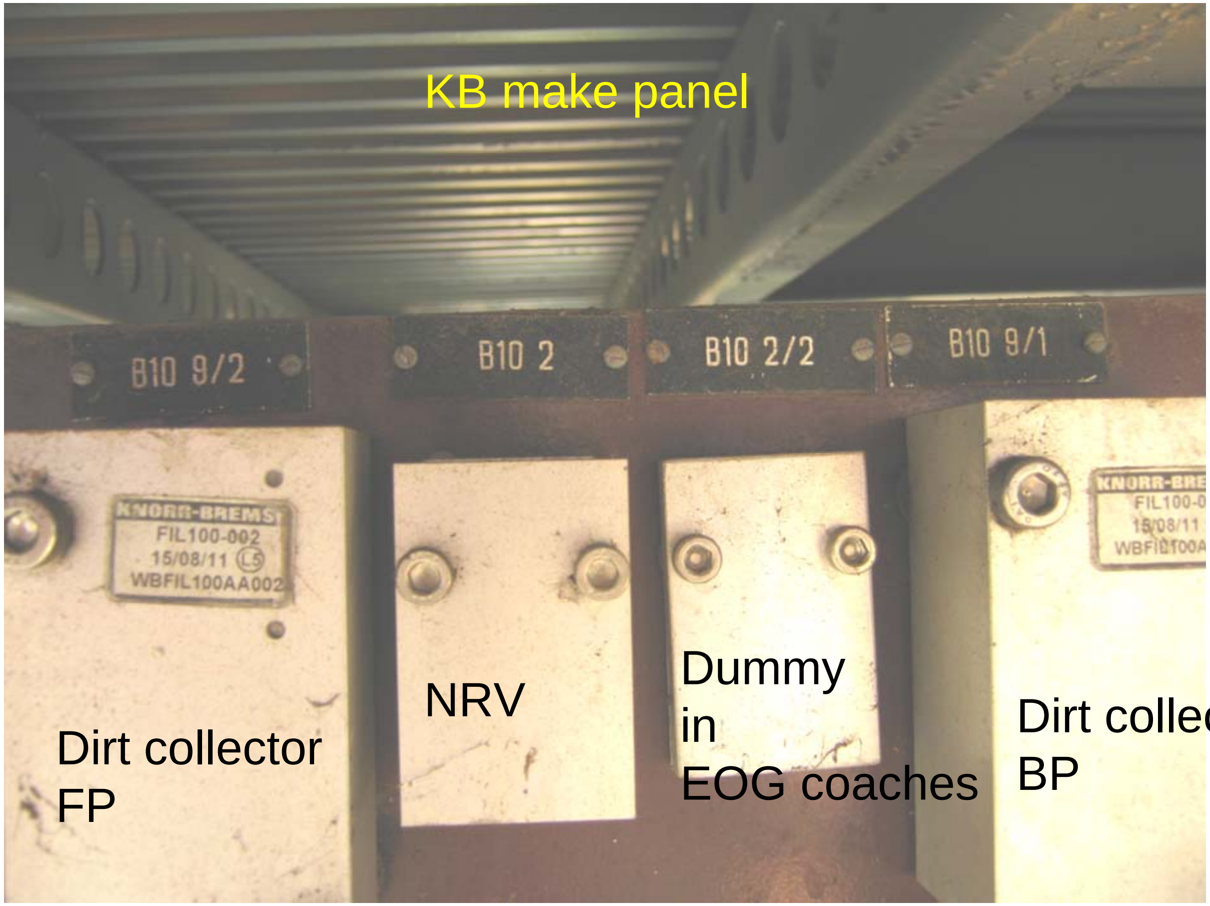


Knorr Bremse make air reservoir

During brake application both cylinder come in to picture



FTIL make Air Reservoir (stainless steel)

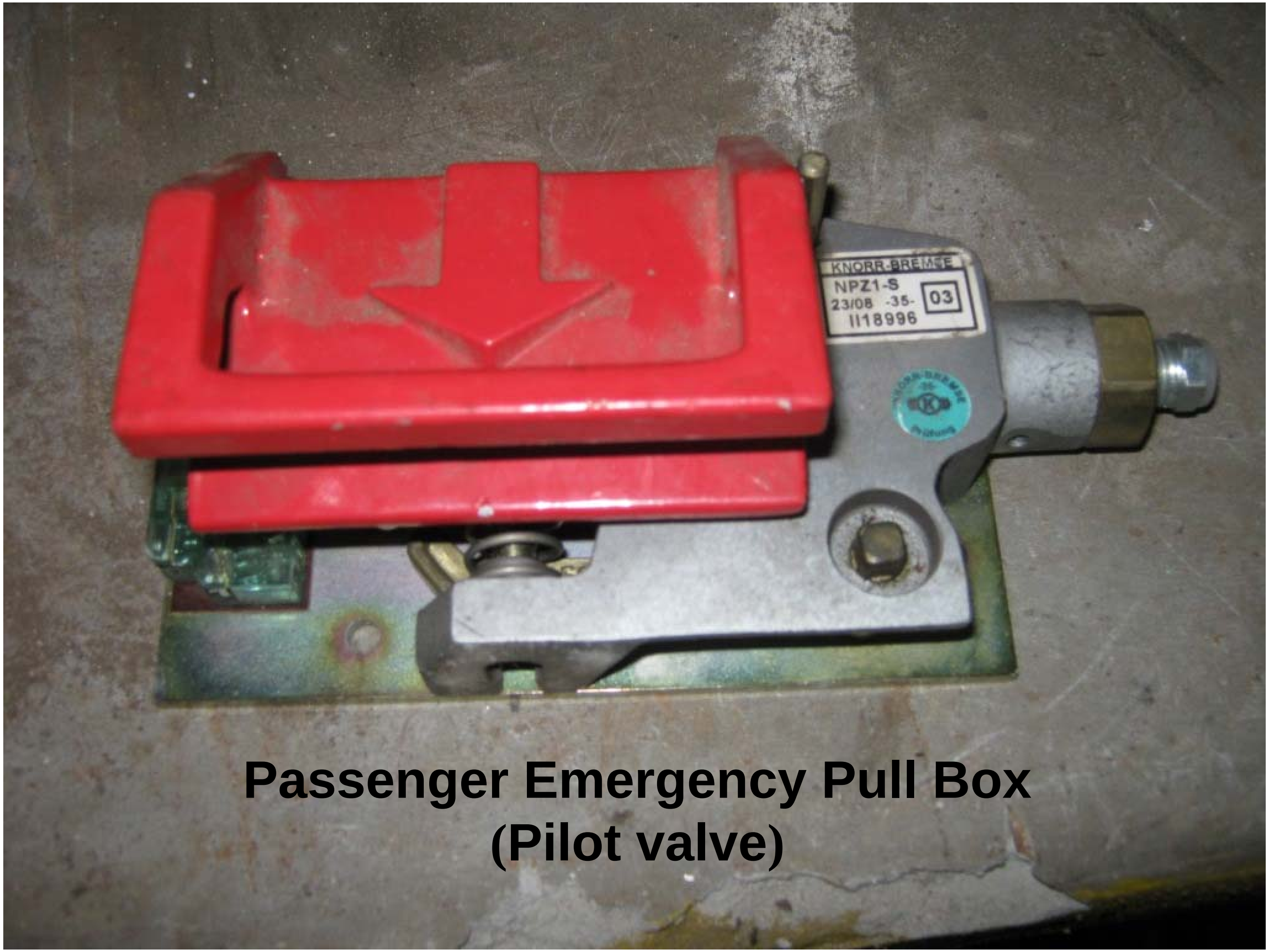


<u>For Dirt collector cleaning following items is required</u>

- Internal circlip plyer
- 6 mm allen key
- Soap solution,
- Screwdriver

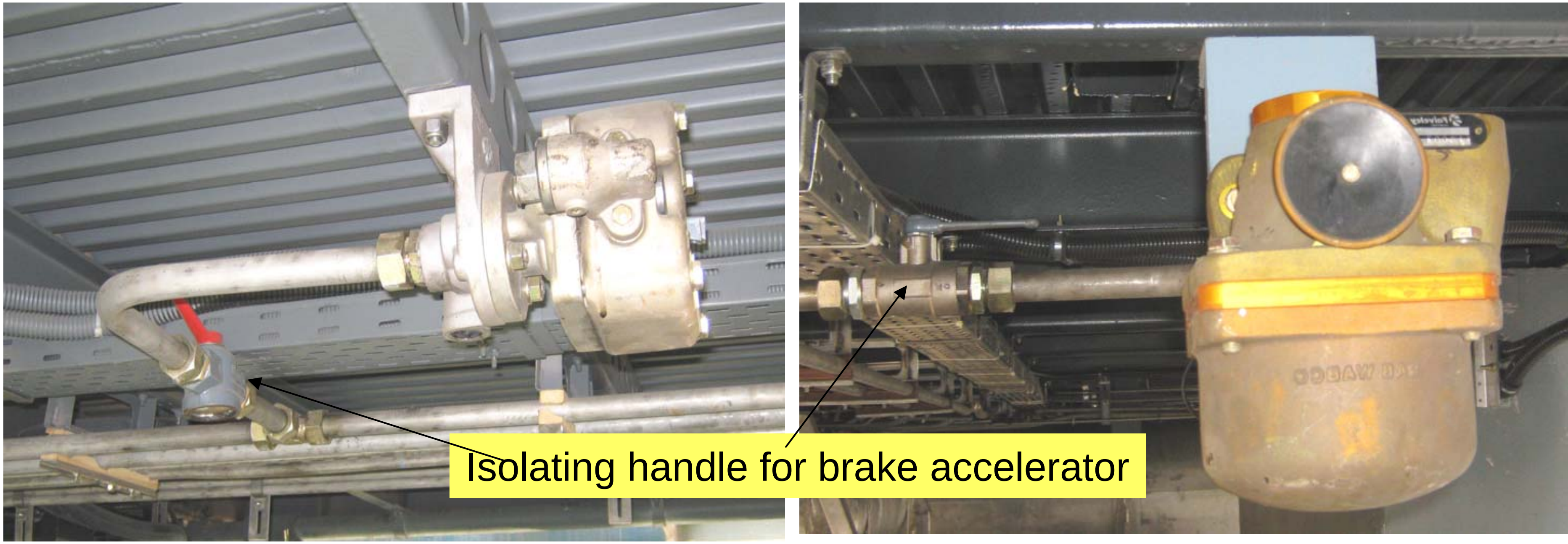
Passenger Emergency Pull Box
(Pilot valve)

Rotate by “rectangular key for setting after chain



Passenger Emergency Pull Box
(Pilot valve)

Emergency Brake valve (PEAV) of KB make



Knorr Bremse Make

FTIL Make

<u>Brake accelerator</u>

Brake accelerator for simultaneous brake application in a rake.

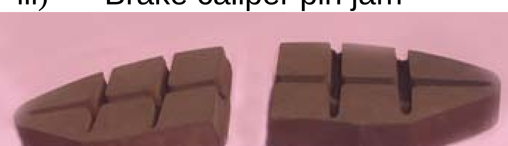
(Direct air release from BP to Atmp. according to rate of fall of BP pressure)

<u>Break Cylinder</u> Size: **10 inch**,
Piston stroke: 21 mm (max),
Slack capacity: 160 mm (min),
BC pressure: 3.0 kg/cm^2,
Its having in build slack adjuster
No Manual adjustment is required to make after Brake pad replacement.

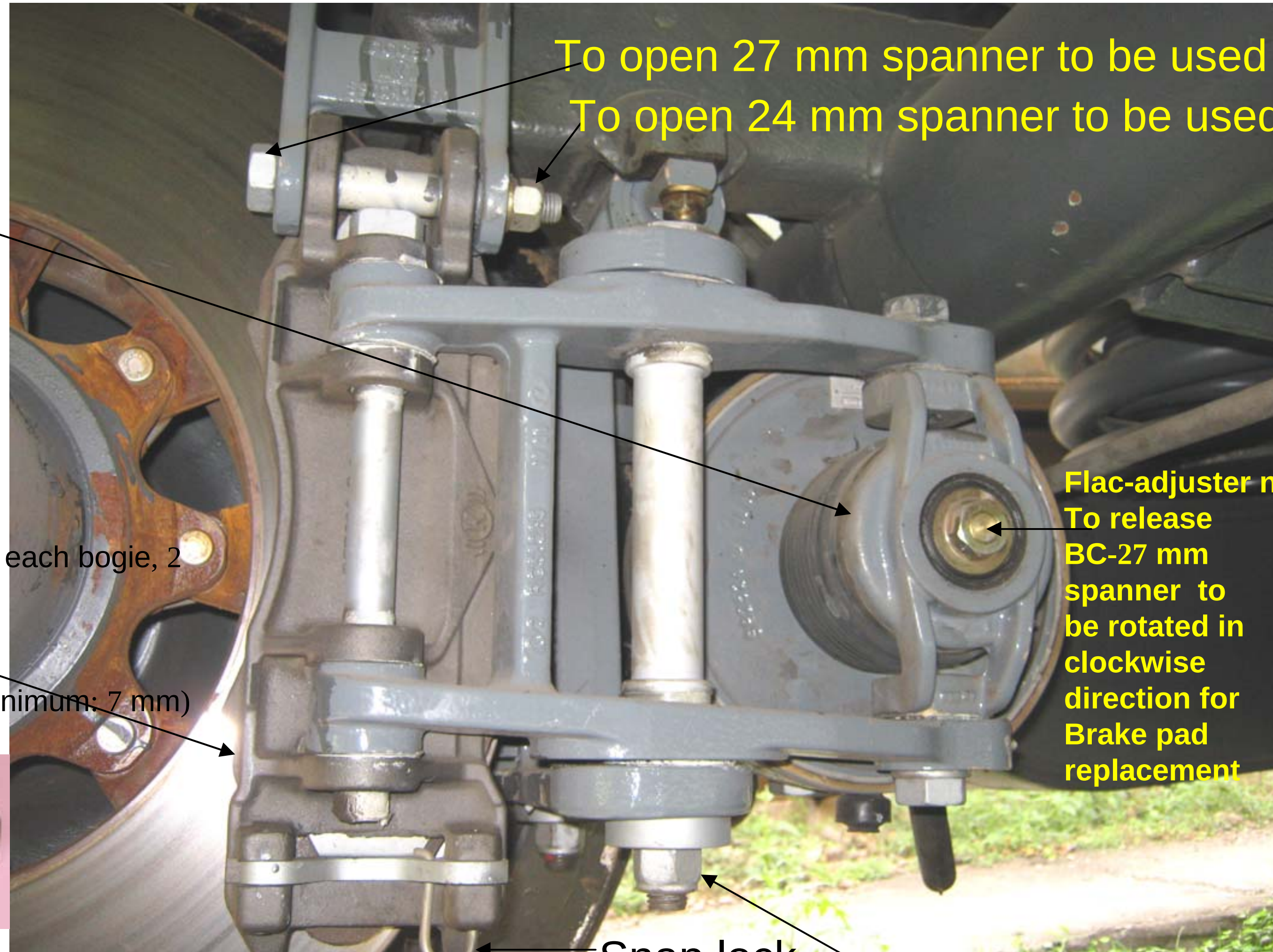
Brake pad (35 mm)-32 nos (16 on each bogie, 2 on each caliper)

<u>Check Points:</u>

i) Snap lock spring tension.
ii) Brake pad(35 mm) wear (minimum: 7 mm)
iii) Brake caliper pin jam



Brake pad

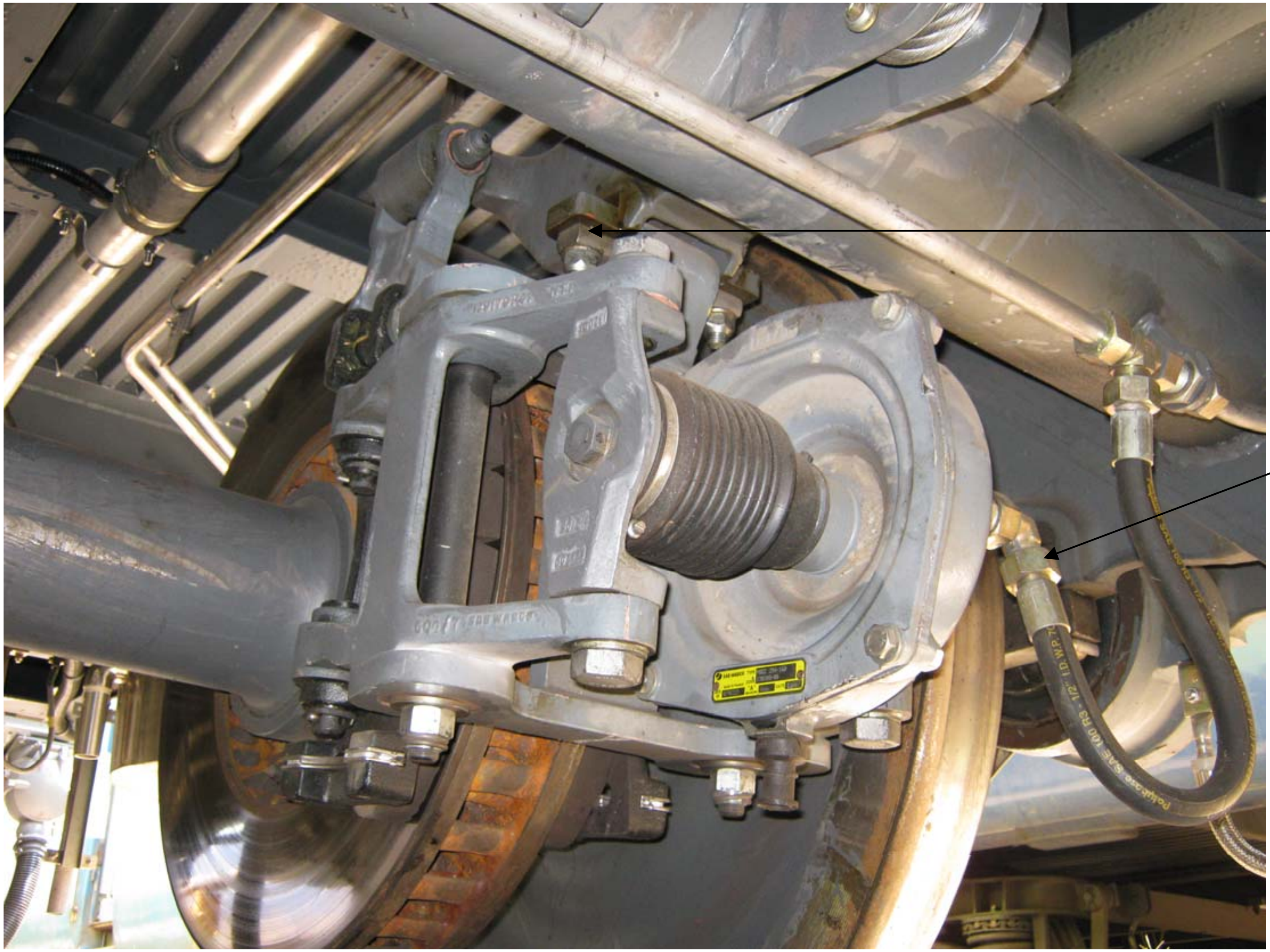


Snap lock

Caliper pivot (Bottom)-
30 mm ring spanner
to be rotated in
counterclockwise direction in
for its free movement

<u>Brake pad manufacture</u>
JURID 877 & BECORIT 984

Brake caliper unit KB make



M16x60,
170N-m

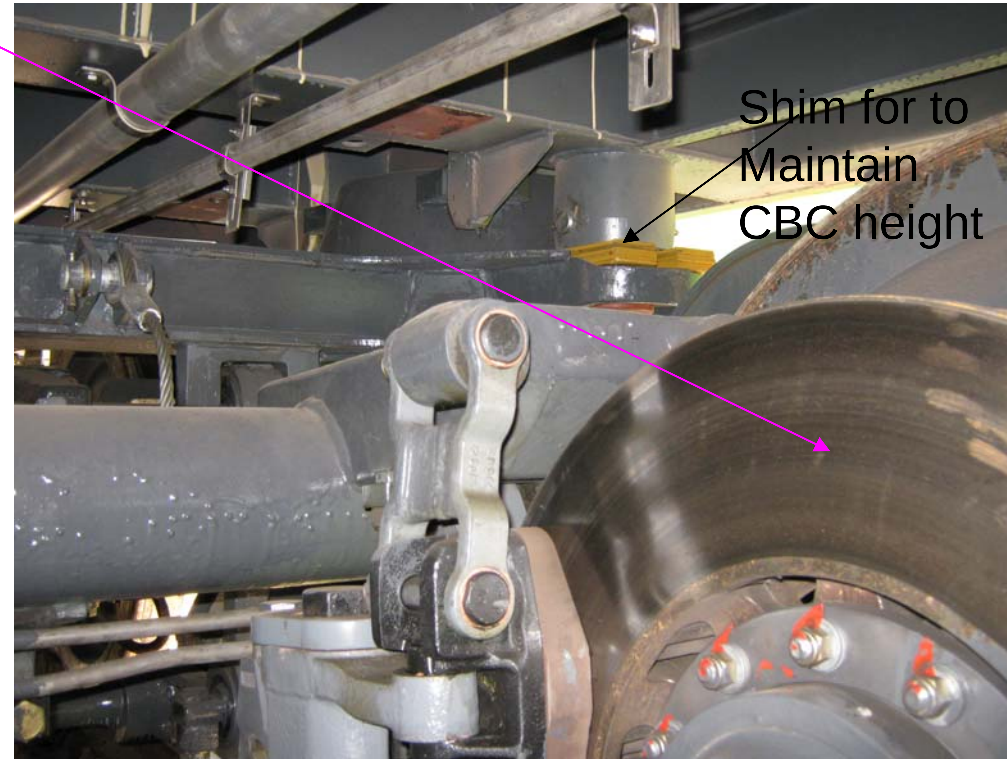
To open BC hose
pipe-32 mm
spanner is
required

FTIL make Brake Caliper unit

Caliper Ratio 1:2.17 for all coaches except power car, which is 1:2.48

Brake Disc (dia-640mm, width-110 mm), Gray cast Iron

Check Points:



•Crackness

•Wearness (7mm in one side, 14 mm in both side)

•Pitting

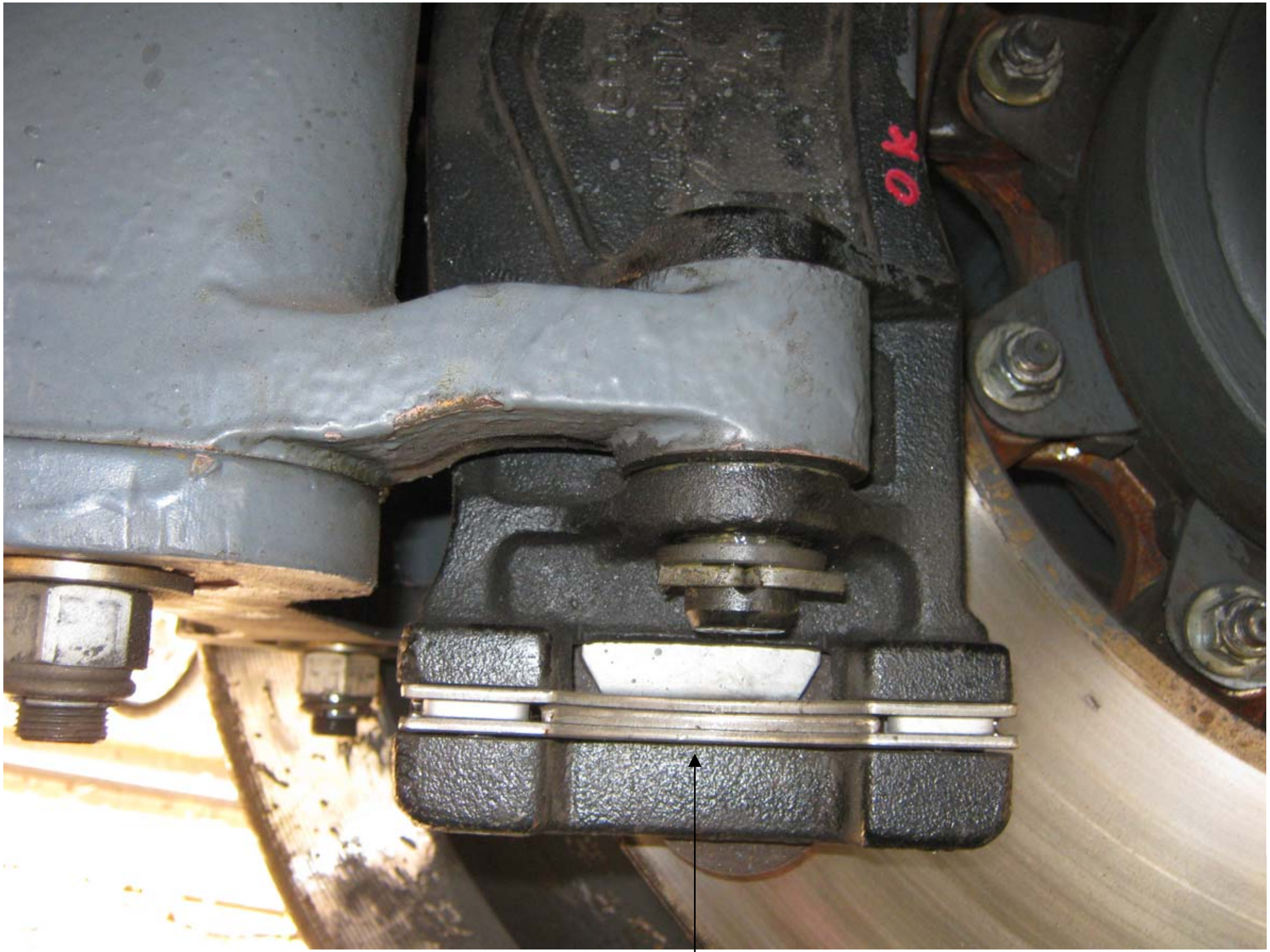
•Ovality

•Heat Crack

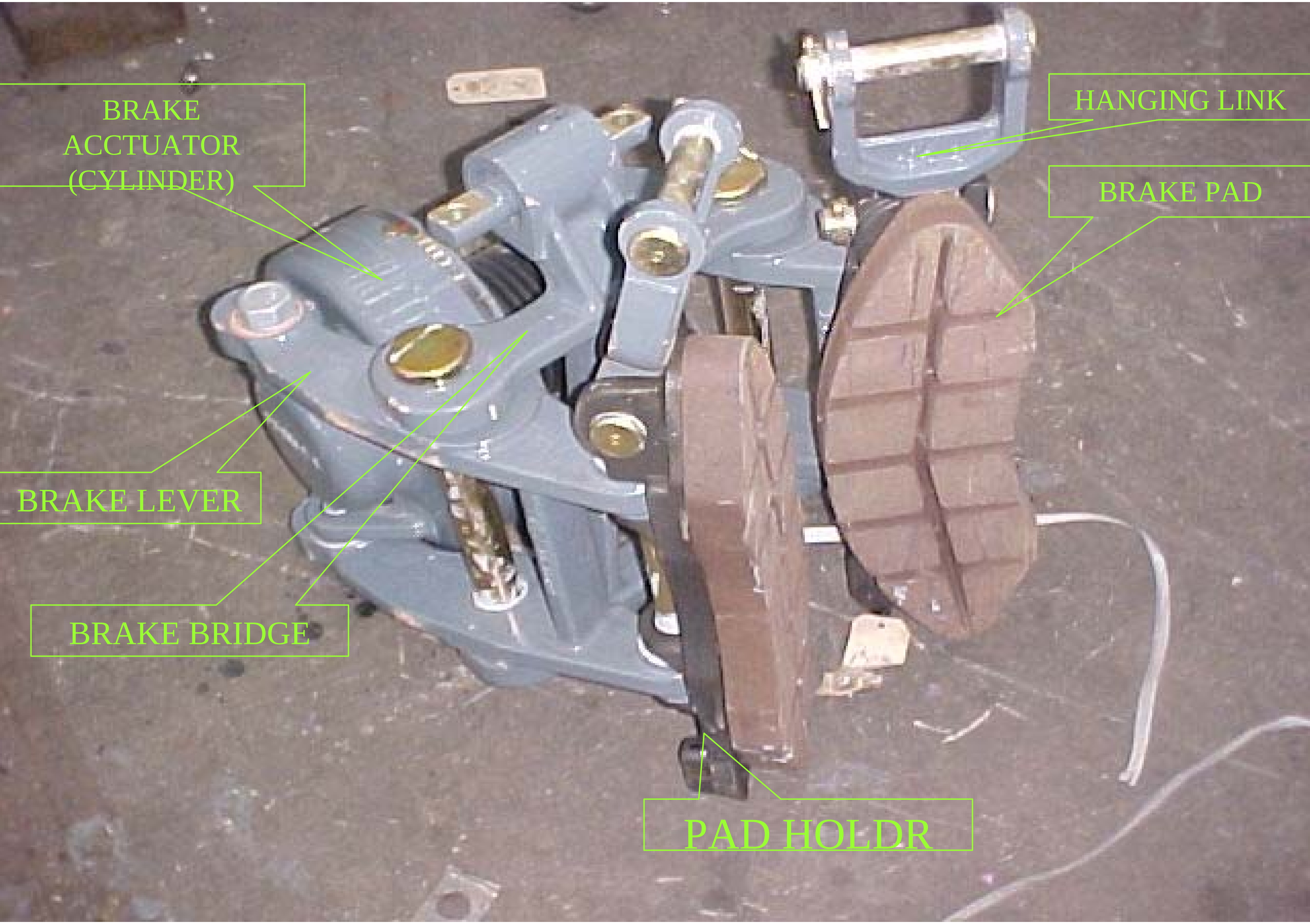
•Thermal Crack

•Breakage of Cooling fins (consecutive 5 nos. & 9 nos. of fins breakage are not permissible in FTIL make, KB make respectively)

•Tightness of Brake disc nuts.



Snap lock in FTIL make brake caliper

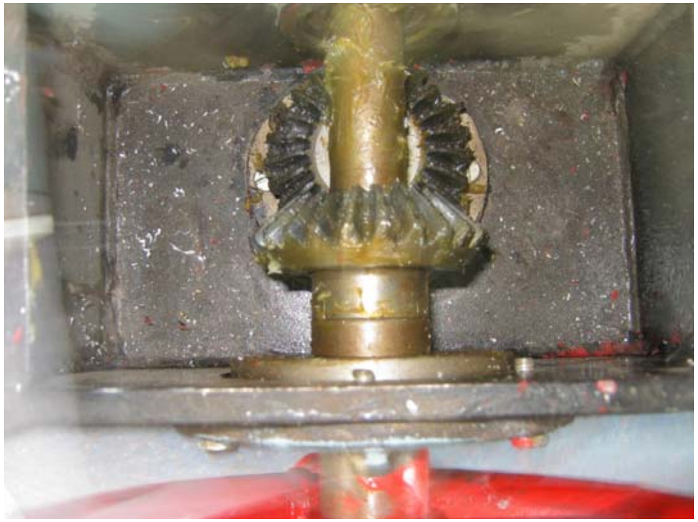

BRAKE ACCTUATOR (CYLINDER)
HANGING LINK
BRAKE PAD
BRAKE LEVER
BRAKE BRIDGE
PAD HOLDR

Bite joint
ferrule

Hand brake arrangement of Power car

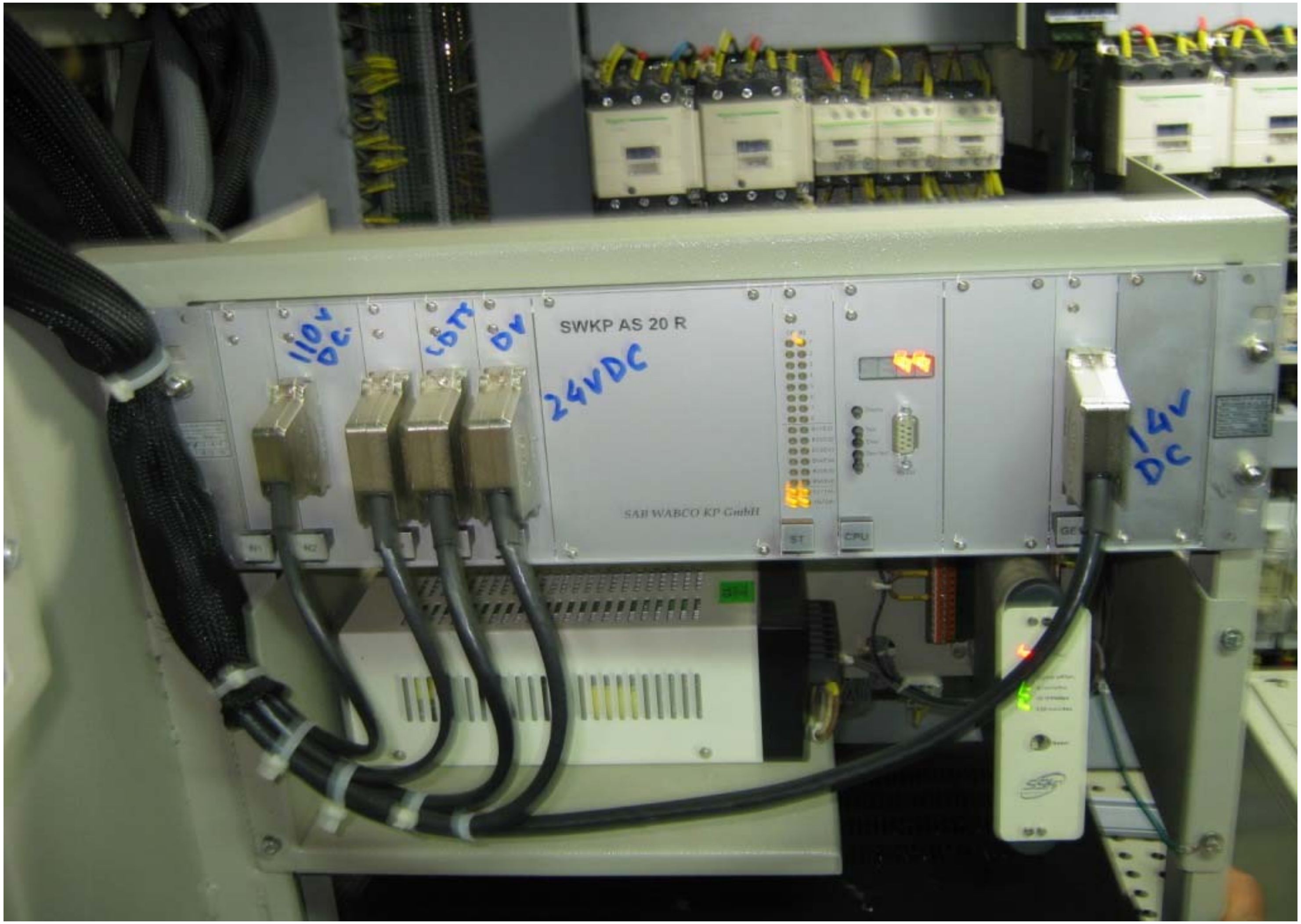
Release condition of Hand Brake

Applied condition of Hand Brake

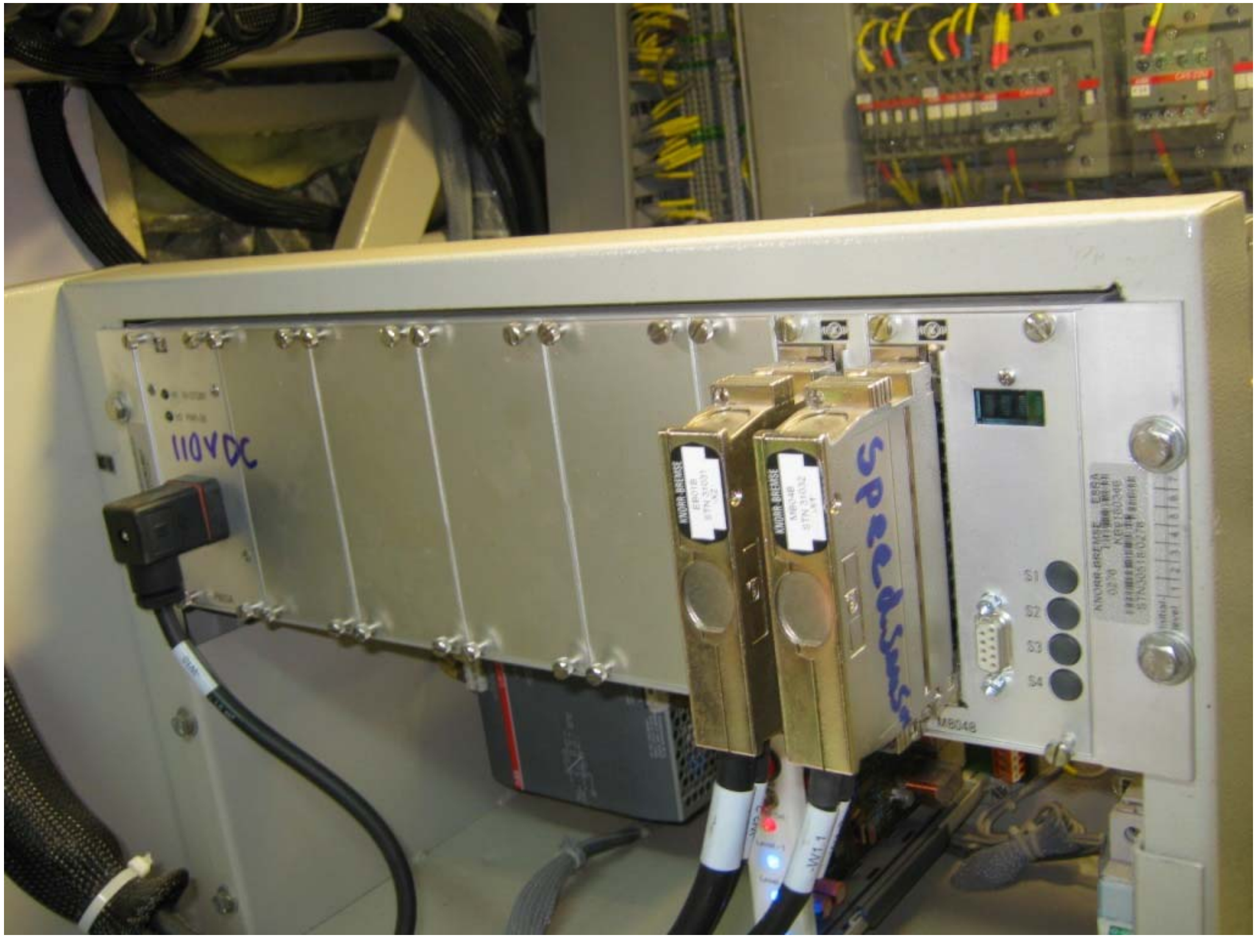


Gear arrangement in in
Guard portion in power car

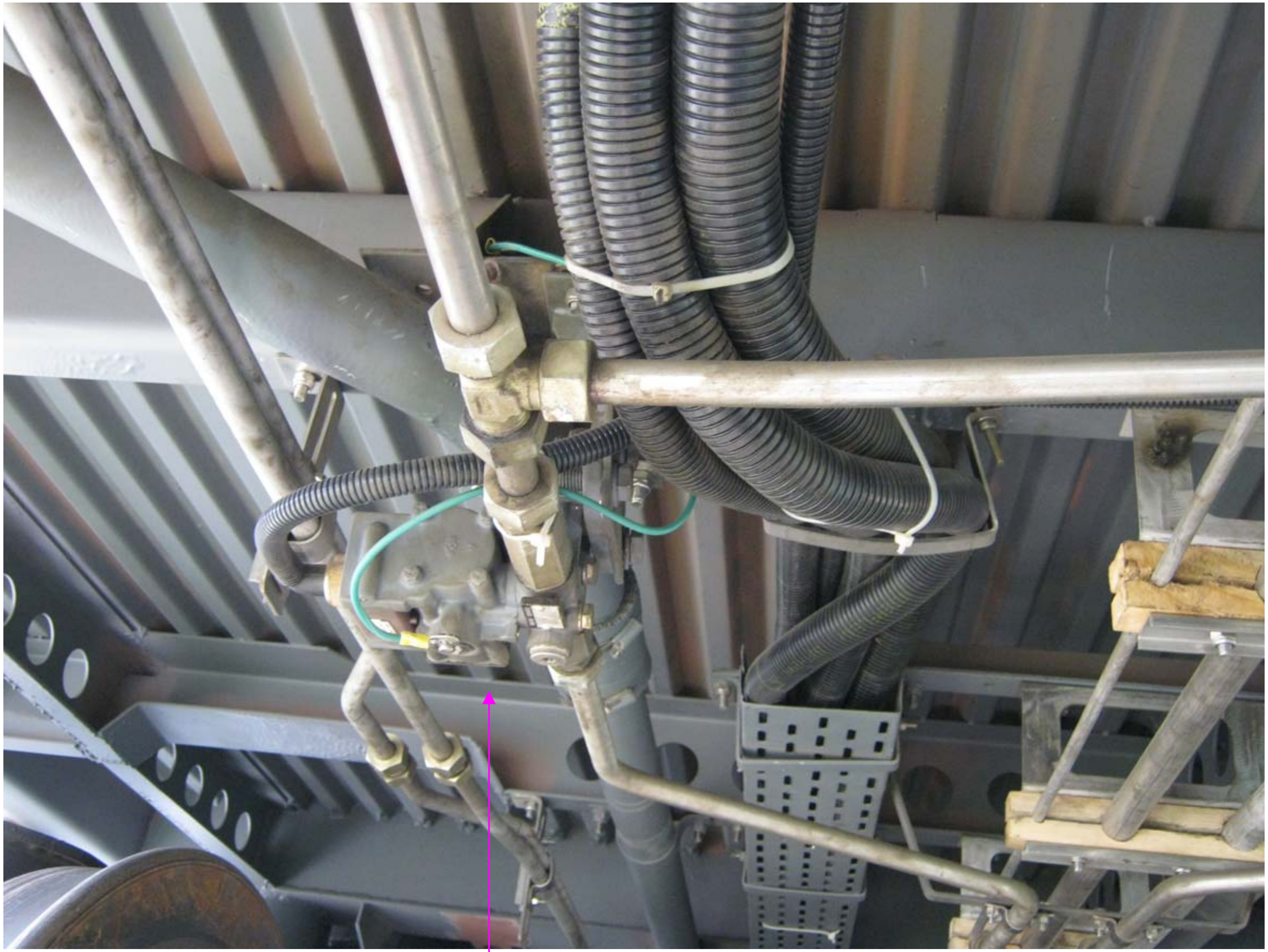
Indicator of hand brake in power car



Microprocessor Control Unit of Knorr Bremse Make



Microprocessor Control Unit of FTIL Make



Anti Skid Valve (Dump Valve)

KB make Speed sensor

Gap between sensor and teeth:

0.7to 1.1 mm for KB & 1 to 2 mm for FTIL


M20 bolt tightened in axle by 200 N-m
M8 X 35 bolt to be tightened by 40 N-m torque
Phonic cowwheel for speed sensor
(80 Teeth)- **fitted for speed sensing**

Testing procedure in KB make

Step-1: Press S2 for testing of dump valve

89➔ testing code

7201➔ failure in one axle

7301➔ failure in more than one axle failure

If display returns back to "99" WSP function is ok.

Step – 2: Press S1 to display failure code

95➔ volatile faults after rectification

Step - 3: Press S3 to clear volatile faults code

AXLE NO.	Knorr Bremes			
	Failure code of speed sensor		Failure code of dram valve	
	OC & SC	Gap in phonic wheel	SC	OC
1	11	12	13	14
2	21	22	23	24
3	31	32	33	34
4	41	42	43	44
99 ➔ WSP function OK OC ➔ indicate open circuit, SC ➔ indicate short circuit				

Testing procedure in FTIL make

Step-1: Press S2 for testing of dump valve

89→ testing code

7201→ failure in one axle

7301→ failure in more than one axle failure

If display returns back to “99” WSP function is ok.

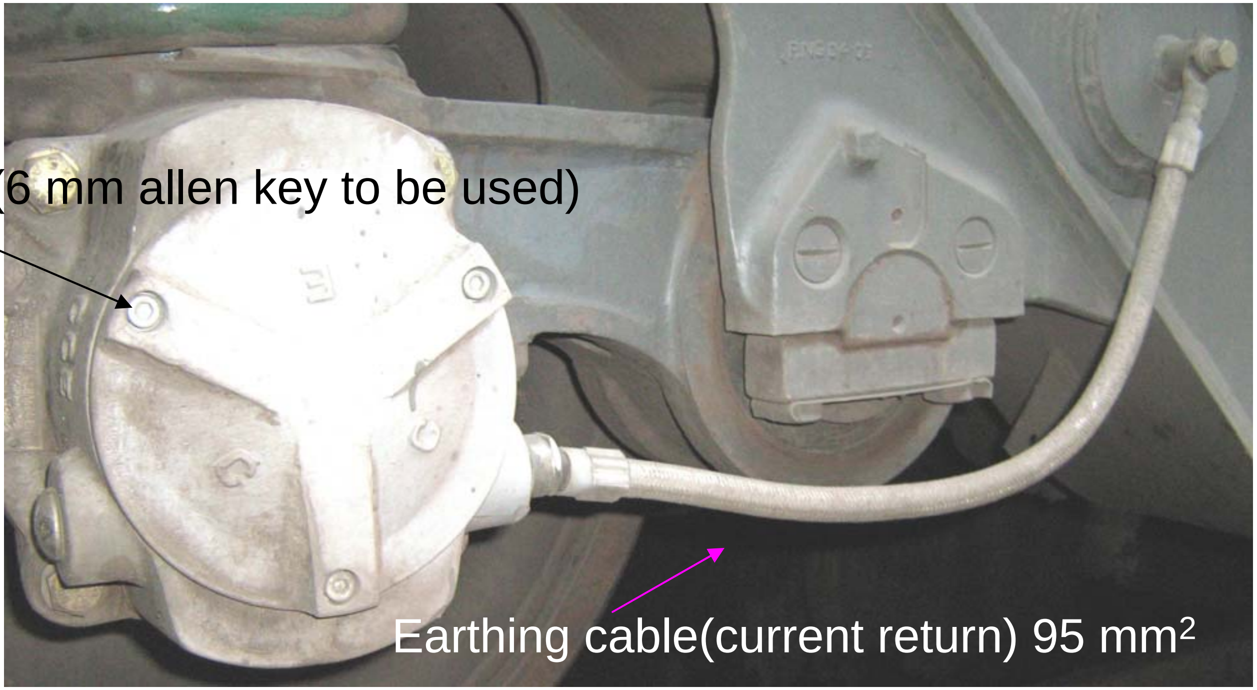
Step – 2: Press S1 to display failure code

95→ volatile faults after rectification

Step - 3: Press S3 to clear volatile faults code

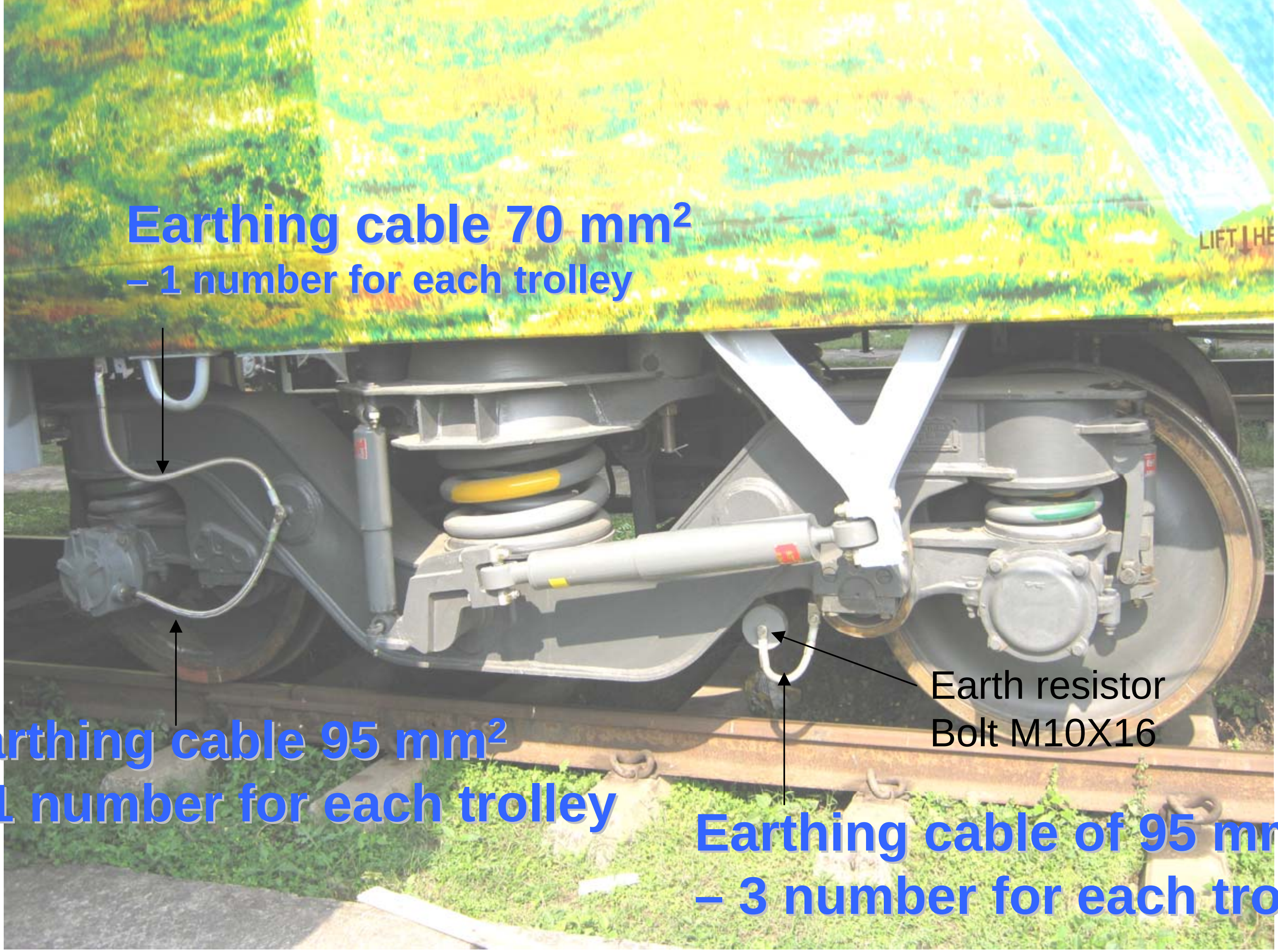
FTIL		
Axle No.	Failure code of speed sensor	Failure code of dram valve
1	11	14
2	21	24
3	31	34
4	41	44
99 → WSP OK		

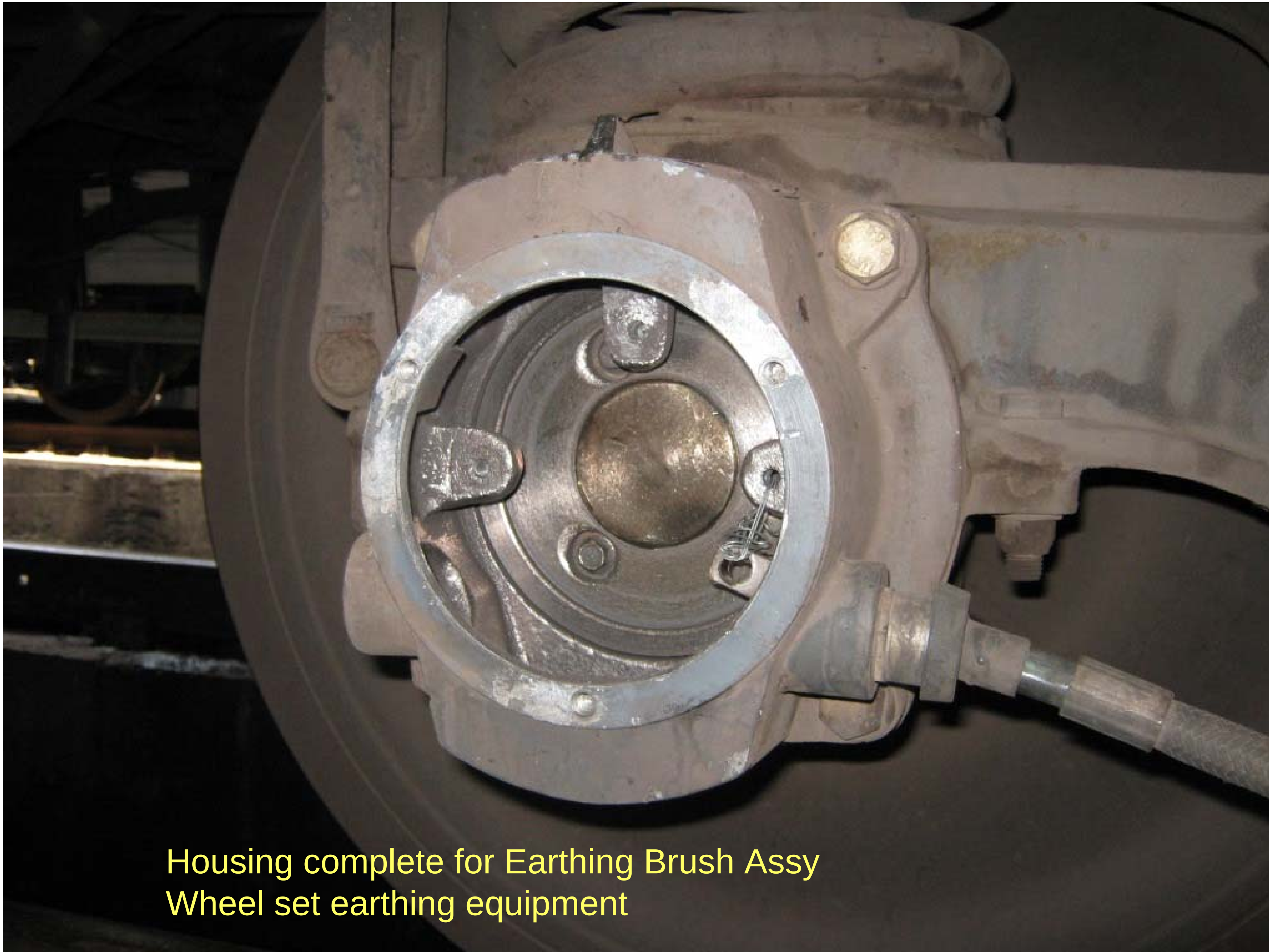
Wheel Set Earthing Equipment



Wheel Set Earthing Equipment
(Available in Wheel no. 1 & 5)

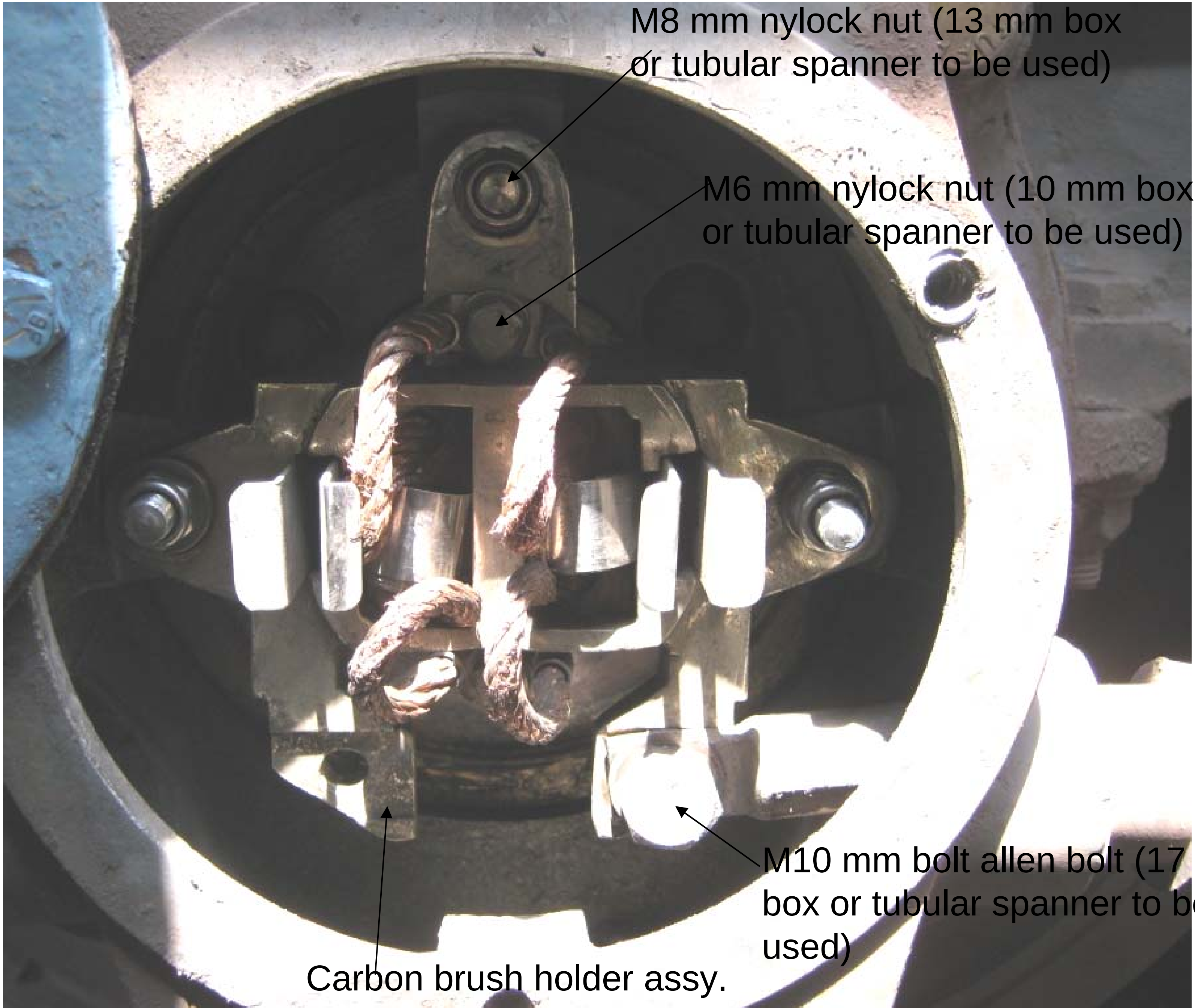
Two manufacturer
1. Darshan Enterprise, New Delhi,
2. Chanda & Chanda, Kolkata


Earthing cable 70 mm²
– 1 number for each trolley
Earth resistor
Bolt M10X16
Earthing cable 95 mm²
– 1 number for each trolley
Earthing cable of 95 mm²
– 3 number for each trolley



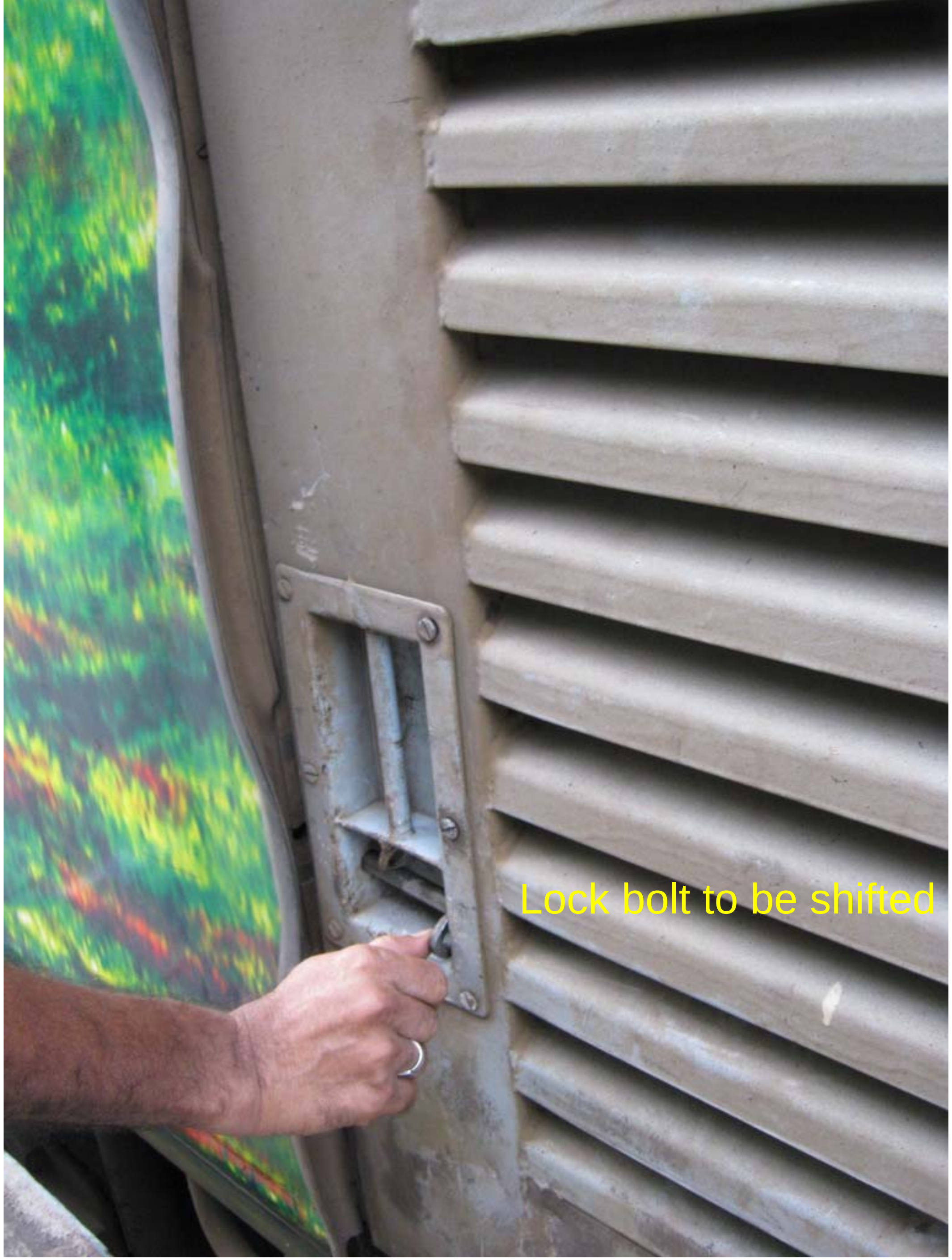
Housing complete for Earthing Brush Assy
Wheel set earthing equipment

Oilier ring (Disc holding the slip assy.) of Wheel set earthing Equipment


M8 mm nylock nut (13 mm box or tubular spanner to be used)
M6 mm nylock nut (10 mm box or tubular spanner to be used)
M10 mm bolt allen bolt (17 mm box or tubular spanner to be used)
Carbon brush holder assy.

Upper gear in LHB coach

Luggage door in LHB power car coach

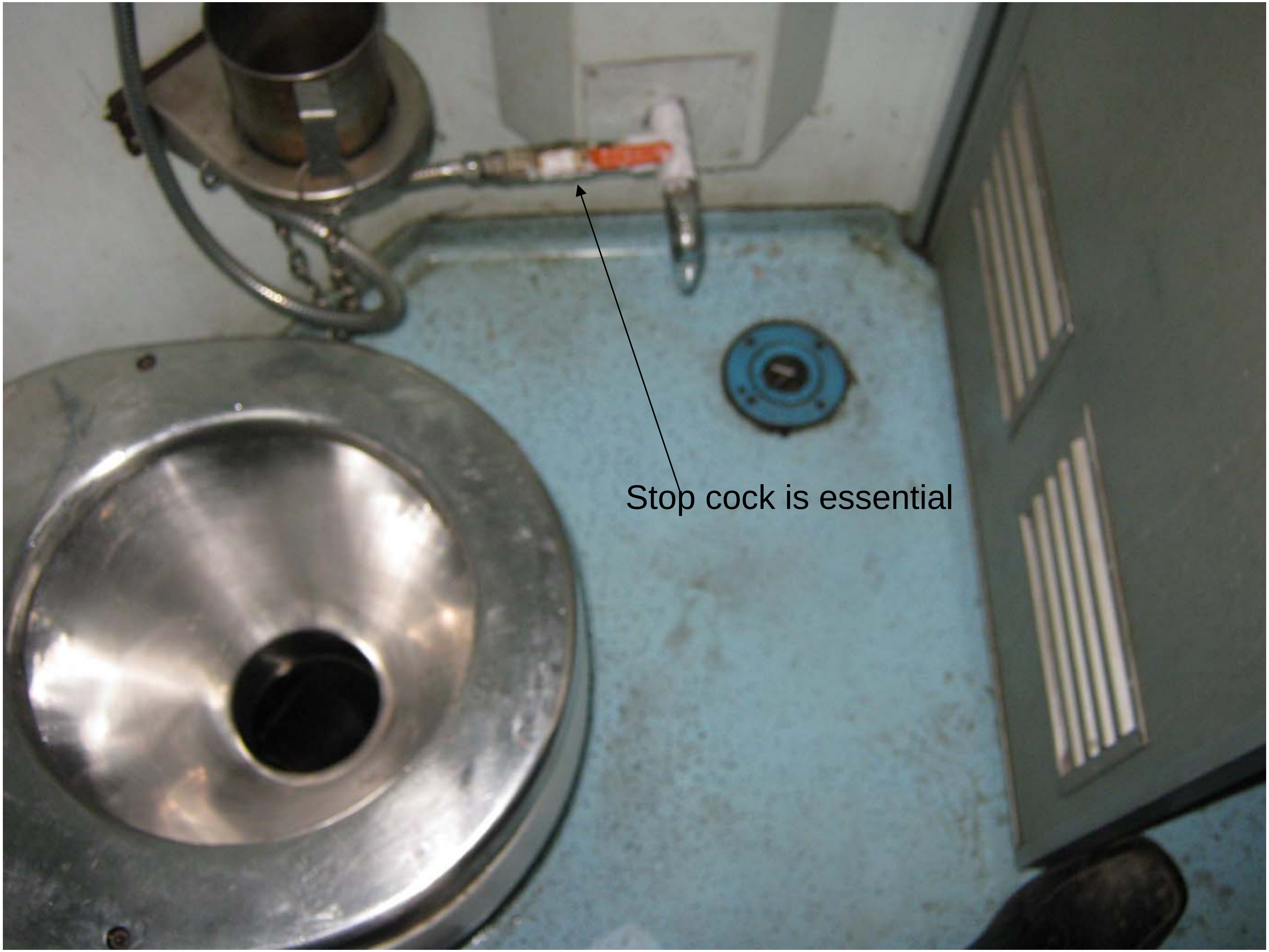


Push the luggage door

Push the from out side

Down the lock and shift the door


No stop cock has been provide
in coaches from RCF


Stop cock is essential


First AC back rest holding clamp

Inside of the toilet

Outside of the toilet

Close door detector

5 mm (dia) x 20 mm (length) X 0.8 mm (pitch),
Alen key size: 3mm

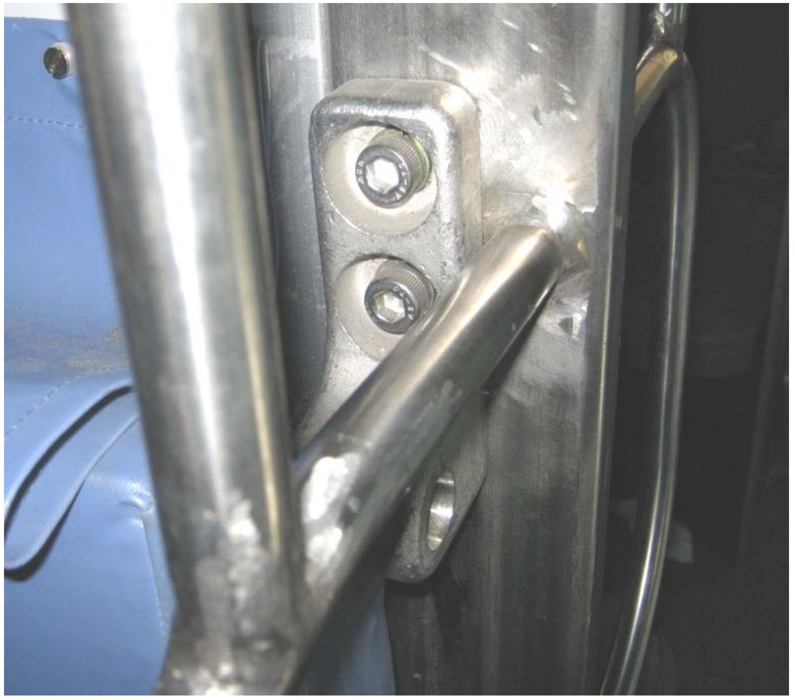
Counter shank headed Alen screw in Close door Detector of size 6 mm (dia) x 16 mm (length) X 1 mm (pitch),

Alen key of size 4mm is required for maintenance

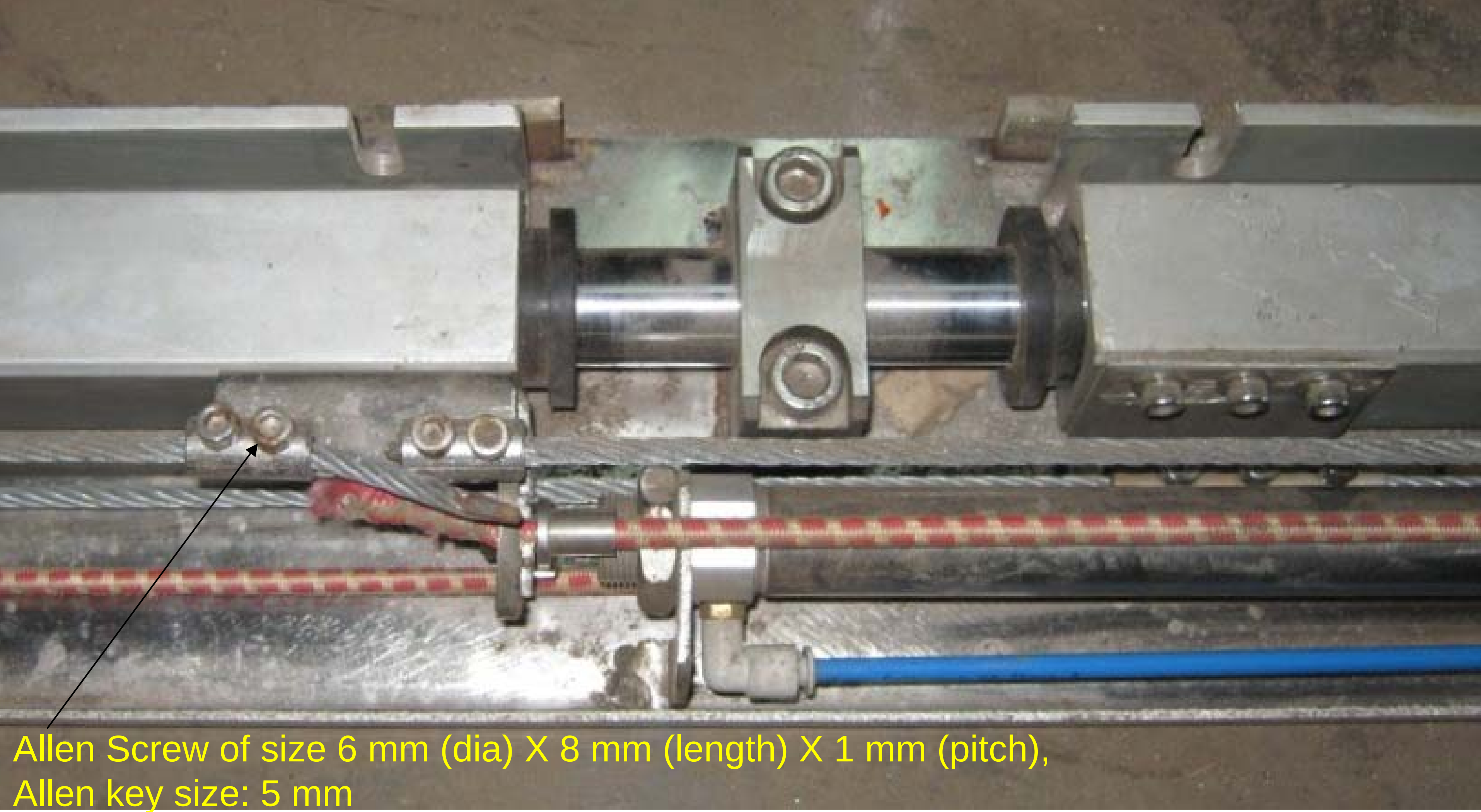
Stainless steel TO catch

SS Toilet door handle

Failure prone second AC safety guard rail in LHB coaches



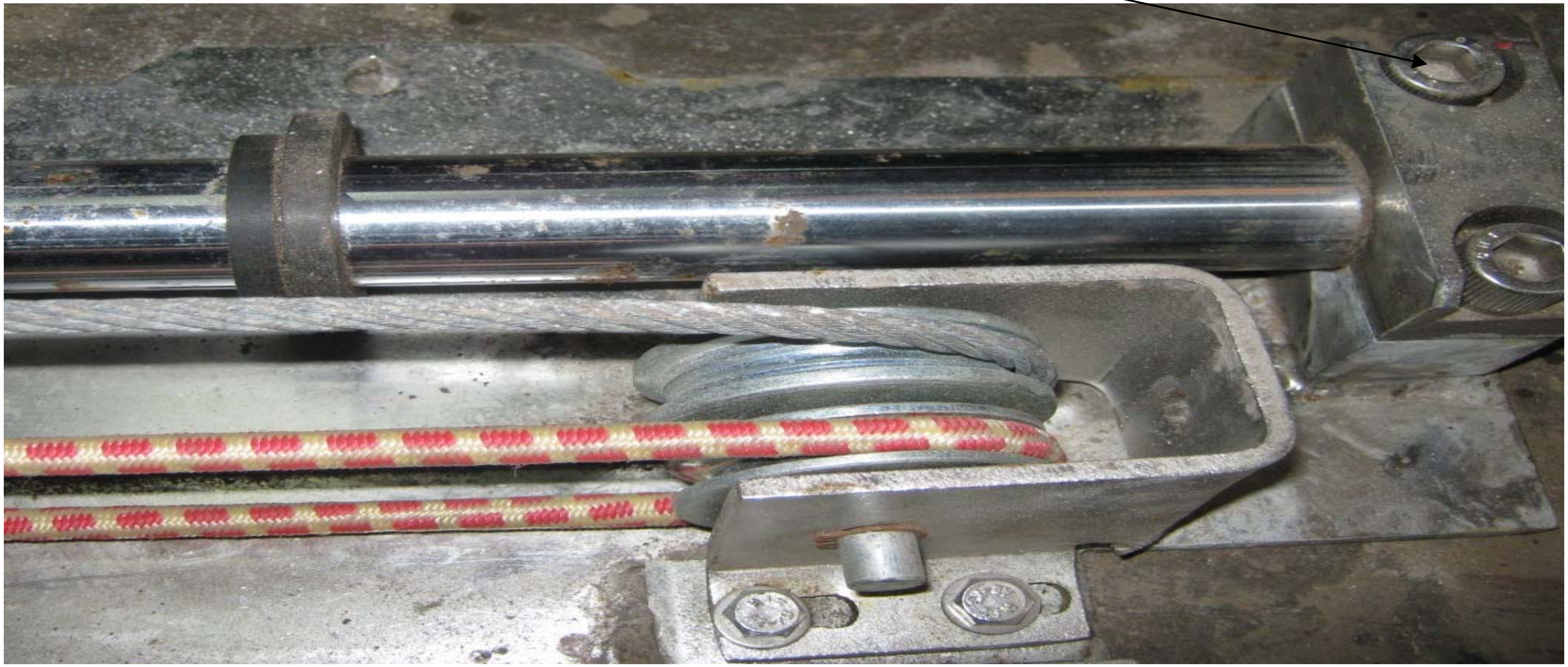
<u>Middle Berth holding bracket</u> at end side
of size 10 mm (dia) x 50 mm (length) X 1.5 mm (pitch),
Allen key of size 8 mm



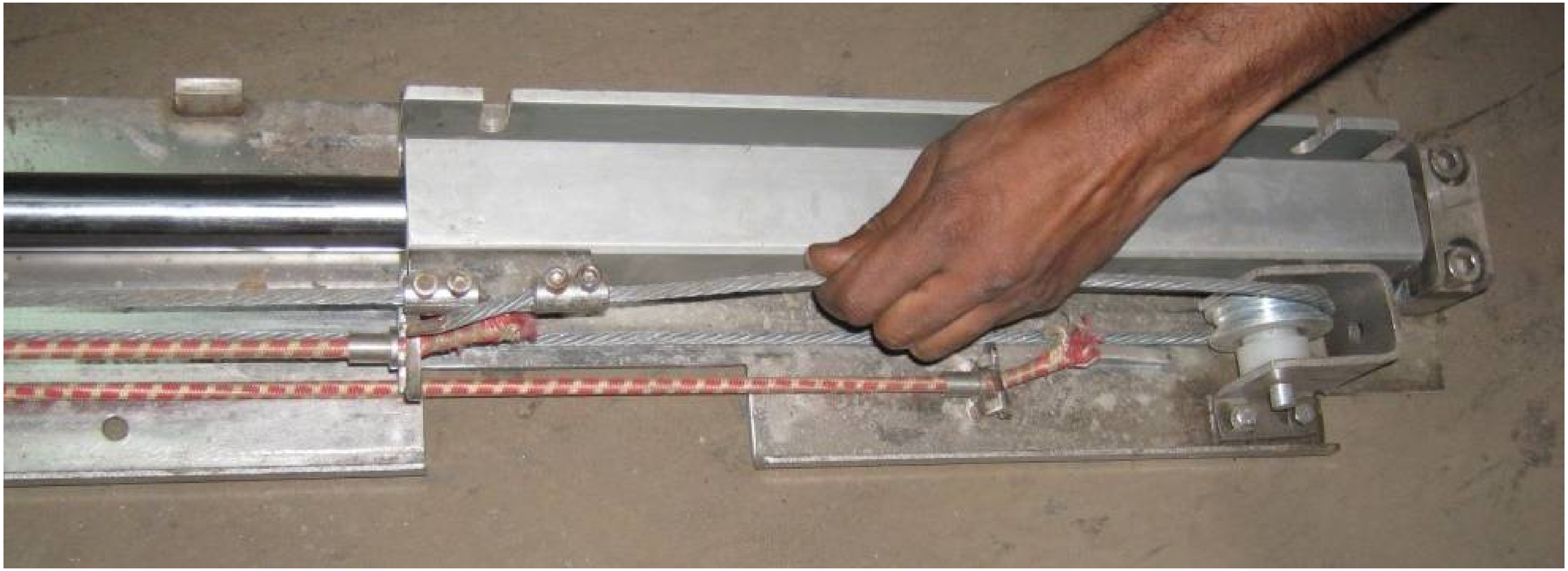
Vestibule sliding door

Sliding door arrangement in **close condition** at middle portion

Alen Screw of size 12 mm (dia) X 40 mm (length) X 1.75 mm (pitch),
Allen key size: 10mm



Sliding door arrangement in close condition rear end



Sliding door arrangement in **open condition**

Coat hook

Counter shank headed Allen screw of size 6 mm (dia) x 16 mm (length) X 1 mm (pitch),

Allen key of size 4mm

Bundle rack

Allen Screw of size 8 mm (dia) x 35 mm (length) X 1.25 mm (pitch),

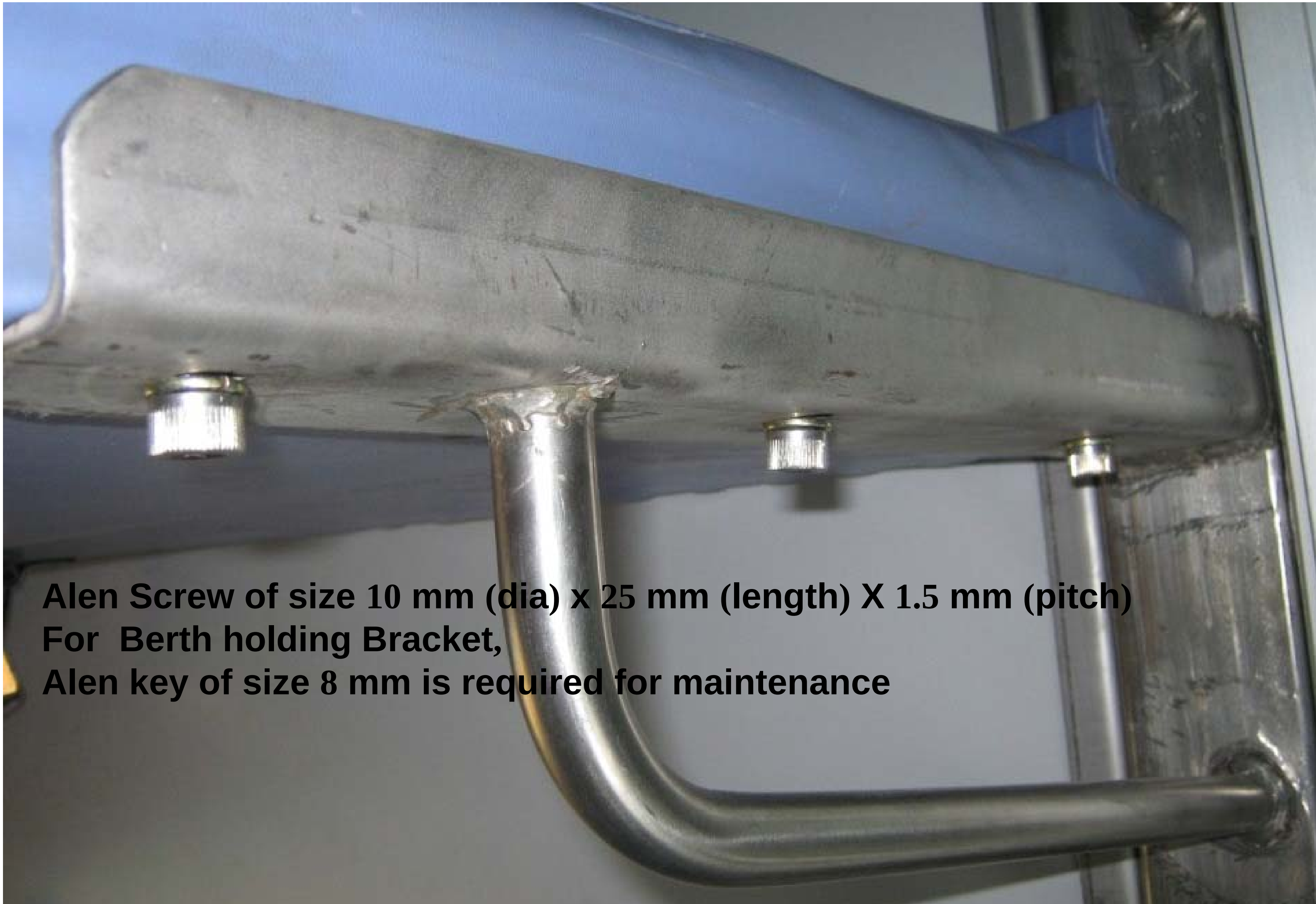
Allen key of size 6 mm

Mirror clamp

Hex headed Grab screw required in mirror bracket of size 4 mm (dia) x 10 mm (length) X 0.7 mm (pitch),
Alen key of size 1.5 mm is required for maintenance

Door Stopper
of size 8 mm (dia) x . mm (length) X 1.25 mm (pitch),

Allen key of size 6 mm


Alen Screw of size 10 mm (dia) x 25 mm (length) X 1.5 mm (pitch)
For Berth holding Bracket,
Alen key of size 8 mm is required for maintenance

AC three tier safety guard rail

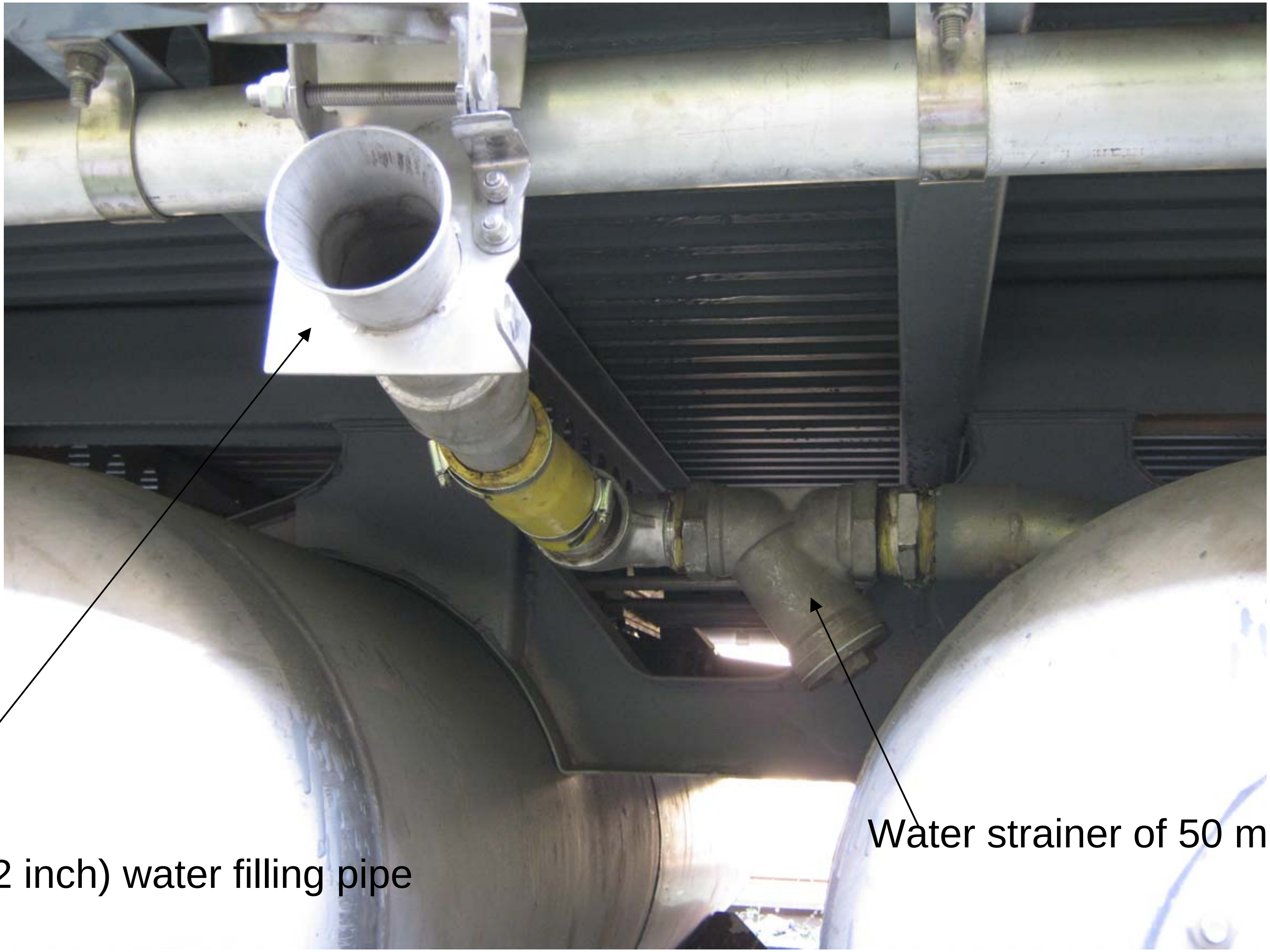

Alen Screw of size 10 mm (dia) X 50 mm (length) X 1.5
mm (pitch) for middle berth holding Chain
Alen key of size 8 mm is required for maintenance


Side Berth holding Bracket,
Alen Screw of size 8 mm (dia) x 25 mm (length) X 1.25 mm (pitch)
Alen key of size 6 mm for maintenance

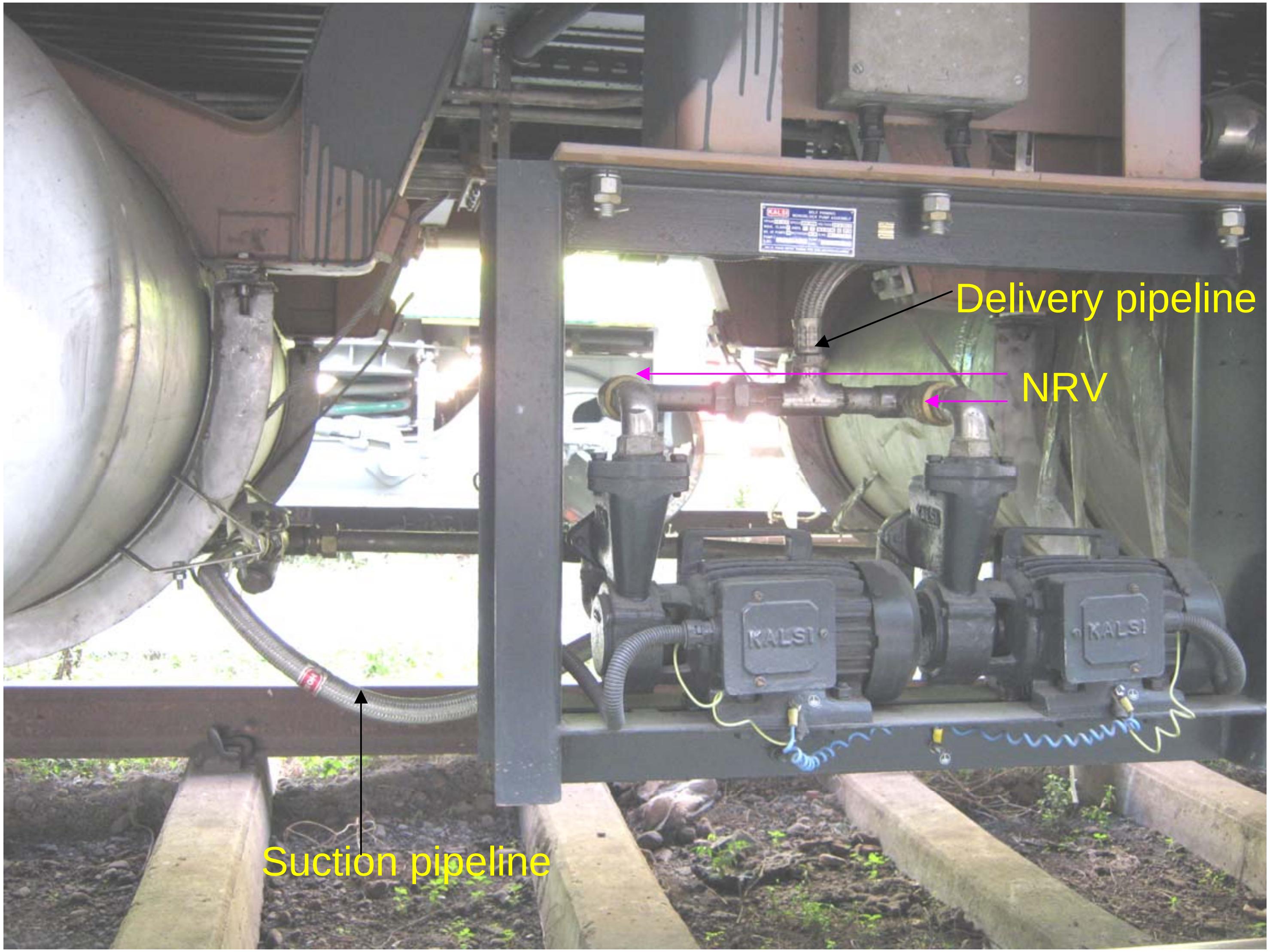
Under Slang Water tank
Capacity

2 numbers x 685 liter & 1 number of 450 liter

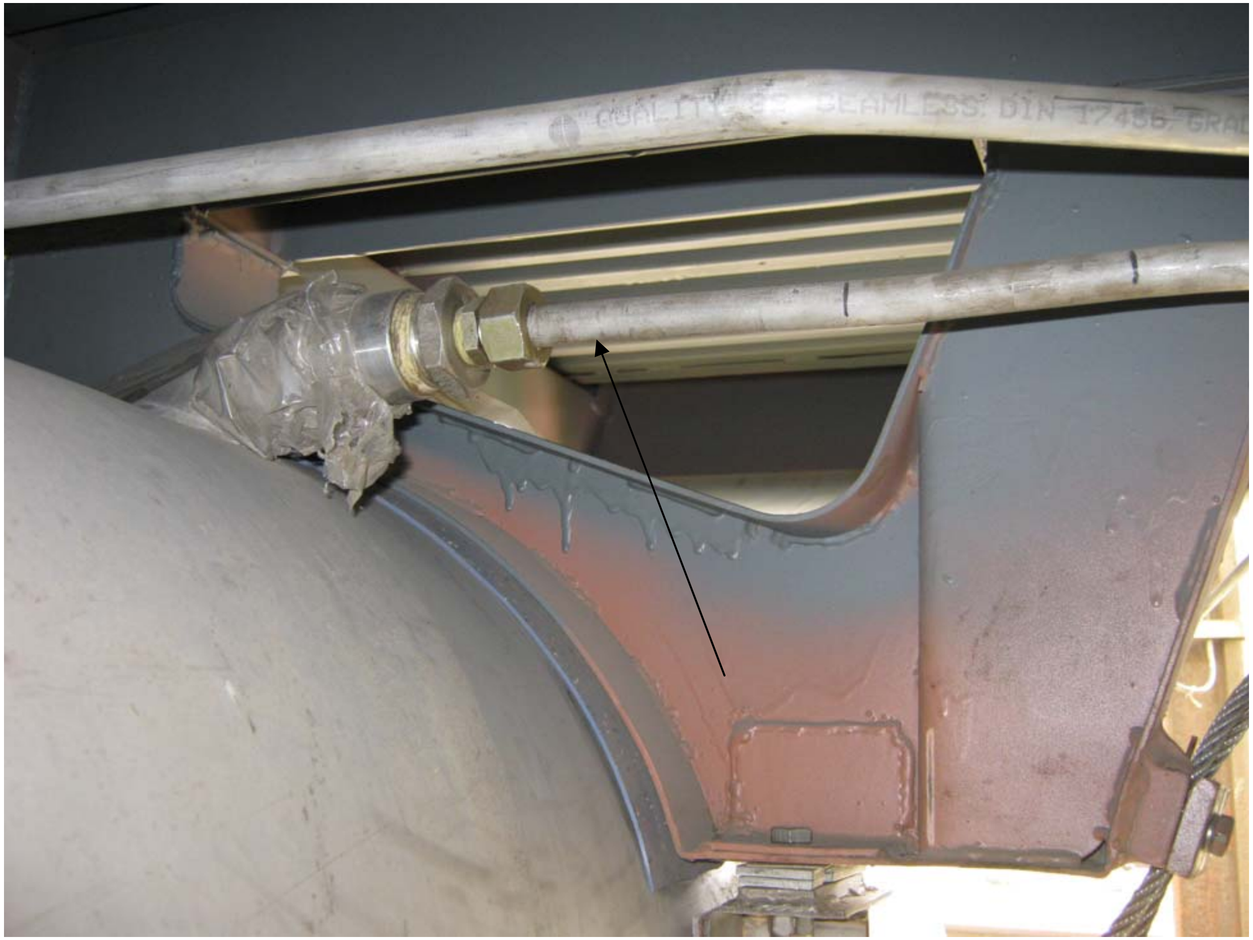
Total of 1820 liter


Water strainer of 50 mm size
50 mm (2 inch) water filling pipe

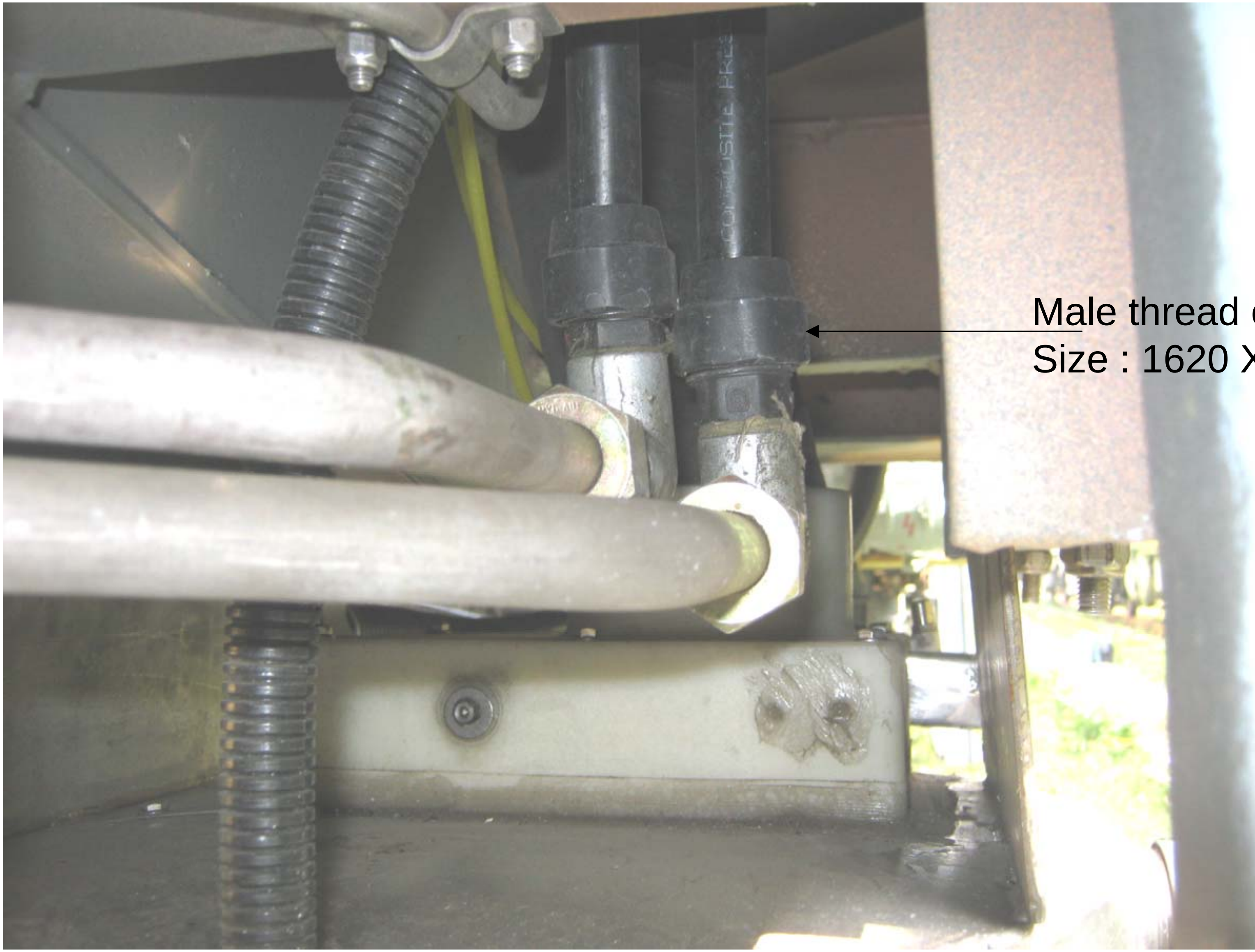
Steel armoured 1 inch flexible pipe



Pump arrangement in LHB coaches



Return pipeline



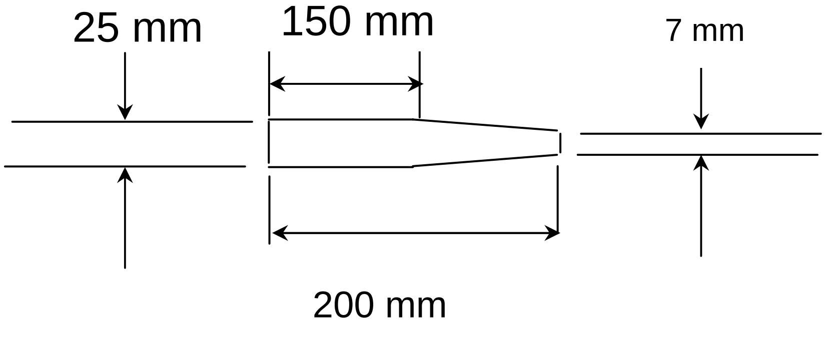
Composite plastic pipeline of M/S Keshra Plast


25 mm ball valve
Water draining arrangement in LHB coach

Gauge for water filling

Tools required for LHB coach maintenance

Chisel of following dimension



8 inch nose plier

10 inch box plier

8 inch Combination plier

Box spanner handle

Tubular Box Spanner

Oil can

Feeler gauge

Torque wrench

Kinfe cutter

Crimping pliers

Allen keys